



SmartSync™ Importers

File Specification

Data Import Guide

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Revision History

Revision	Date	Description
1.6	11/3/2025	In the Import Schedule Changes section, updated description to state that if you are creating future activities using this import, it can take 20 minutes or more to process the files (WEMDOC-7844).
1.5	4/30/2025	- Updated external document references - Limitation on importHistory by Custom Activity for distributed systems is removed by scalable swxiface.
1.4	1/28/2025	Updated external document references.
1.3	9/18/2024	- In Import Schedule Changes Command Line Parameters: - The description of the d<directory> was changed. - Import Schedule Changes command line examples were updated (WFM-142764). - In Import Schedule Changes XML Record Specifications, the irrelevant maximum length limit for the vacationYear field was removed (WFM-141911).
1.2	5/15/2024	In the Import Forecast section, the Import Forecast DTD has been updated and the contactsActive field was added.
1.1	4/26/2024	A new Import History command line parameter (-AC) is introduced to enable and process the ActiveContact field in the Import History file. (WFM-144052)
1.0	3/1/2024	Initial 8.x document - Corrected case-sensitivity of “importhistory” command. - In Import History section, updated importhistory SmartSync spec to import Active Contacts column for TTI Analytics. - In Import History section, the field name has been updated to Active Contacts. (WFM-134897) - In the Import Agent File Contents – Record Specifications section, updated some email related fields, corrected some descriptions. Also in the Import Agent XML Input File Examples section, updated email fields to show valid examples.

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1. Overview

SmartSync provides several predefined imports. Each import has a unique set of values allowing you to choose the values you want to import. Create XML or text input files based on the import file specifications for each import type.

SmartSync imports provide the ability to:

- Pre-populate up to three years of your contact history.
- Pre-populate agent definitions, or continually update agent definitions, from an HR or competing workforce management system.
- Automatically update a schedule based on an agent's call into an IVR system. For instance, calling in late or absent.
- Import contact type (CT) history from a multimedia server, chat, email, fax, and so on.
- Automatically update forecast values using SAS, Excel, and so on.
- Synchronize skill updates from an ACD to WFM.

1.1 Scope of this Guide

Use this document as a reference for using SmartSync importers with NICE WFM. These instructions support WFM Release 8.x and later.

This guide covers these main topics:

- File specifications for importing data into the WFM database.
- The DTD (Document Type Definition) for each import type.
- Input file examples for each import type.
- Command line parameters to run each import type.
- Processing information for each import type.

Note: This guide does not include implementation details to automate importing data.

1.2 Acronym Conventions

Acronym	Complete Name
ACD	Automatic Call Distributor
AHT	Average Handle Time
ASA	Average Speed of Answer
BU	Business Unit
CT	Contact Type
DCM	Digital Channel Management

DTD	Document Type Definition
ID	Identification
MU	Management Unit
OT	Overtime
RCP	WFM Client (Rich Client Platform)
SL	Service Level
WFM	Workforce Management
XML	Extensible Markup Language

1.3 Data Import Types

SmartSync provides these imports:

- Import Agent
- Import Forecast
- Import History
- Import MU Opens
- Import Schedule Changes
- Import Vacation Summary

1.4 Data Import Process

Create the import XML file according to the specifications in this document for each import. To locate and process the import file, you must provide a pathway between the WFM server and the file. For hosted/cloud environments, NICE verifies the EFT process and delivery path.

Using one of these methods, place the file in a specified directory on the WFM server. See [3.1 Import File Name and Location on page 14](#) for the recommended import file location.

You can configure the import program to run automatically at pre-determined (user-defined) intervals or invoke the program manually. Contact NICE Professional Services for assistance.

1.4.1 Data Import Workflow

This step-by-step process assists you in running the import process.

1. Create the import input file.
2. Copy the file to the specified directory. See [3.1 Import File Name and Location on page 14](#).

Example:

```
/totalview/ftp/imports/customer<custID>/agents/agents.1.xml
```

3. Run the import program with the desired parameters.

Example:

```
importAgent -c<custID> -
```

```
f/totalview/ftp/imports/customer<custID>/agents/new/agents.1.xml -i >  
/totalview/ftp/imports/customer<custID>/agents/new/run.log
```

4. Review the **.OUT** or **.ERR** files and correct any errors.
5. Remove the **xmlns** attribute from the **agentList** element and the operation attribute from the **newAgent** or **changedAgent** elements in the **.OUT** file.
6. Rerun the import process with the contents of the **.OUT** file or **.ERR** file.

2. Information for the Import Programs

2.1 Field Length

All entity names and user defined data items are 64 characters unless otherwise noted.

2.2 Locking

The importer reads the existing data from the database without locking it, makes changes, and saves. If the data is modified elsewhere between the time the importer reads the data and the time it saves the changes, an error occurs and the data is not saved.

When an importer modifies data currently being shown to a user in a WFM view and the user tries to modify the same data without re-retrieving, an error occurs when saving the changes and a message is shown saying someone else modified the data. The user needs to re-retrieve the data and make the changes again. This is true whether the importer makes changes or another user makes changes to the data shown on the view.

2.3 Error Messages

Error messages are written as **messages** elements in the output files. The message child is at the top of the element that caused the error.

Example:

```
<messages>Import failed for some reason</messages>
```

.ERR file example:

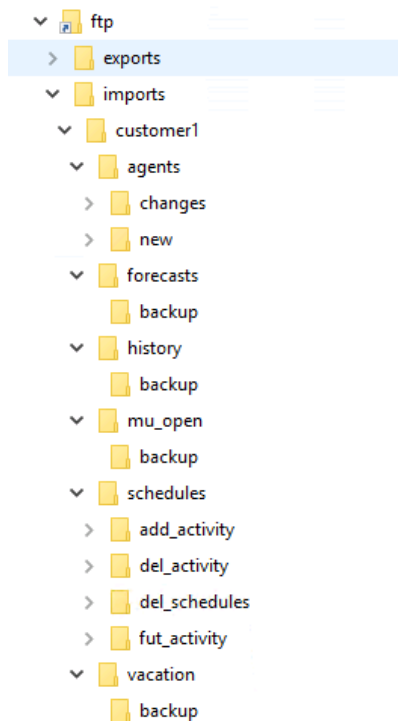
```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<agentList xmlns:ns2="com.iex.tv.smartsync.ws.agentimport">
  <newAgent operation="N">
    <status>Failure</status>
    <messages>Acd not found: 12.</messages>
    <firstName>Ralph</firstName>
    <lastName>Morley</lastName>
    <mu>805</mu>
    ...
  </newAgent>
</agentList>
```

3. Import File Specifications

3.1 Import File Name and Location

To avoid potential problems with processing input/output files, it is recommended you limit the input/output file name to 30 characters or less.

Place the file in the FTP imports folder that follows this folder structure for the import type:



Each new file written to the directory overwrites any existing file of the same name.

In the command line, use the parameter specified for each import type to specify the path and file name.

The import file is not deleted after the import completes.

3.2 Import File Contents – General Specifications

The import file format varies depending on the import type. These formats are supported:

- XML
 - Import Agent
 - Import Forecast
 - Import MU Opens
 - Import Schedule Changes
 - Import Vacation Summary

- Pipe-delimited
 - Import Schedule Changes
- Fields delimited by double quotes and separated by commas
 - Import History

3.2.1 XML Format

The predefined DTD files, which are unique for each import type, specify the elements to include in the import file.

Note: Do not create a DTD file for the import; only use it as a template for building XML input files.

The import file must be a valid, well-formed XML document, based on XML standards. You must follow all standard rules:

- The file must begin with an XML declaration: `<?xml version="1.0"?>` If the database contains international characters, this line must contain an encoding statement. For example: `<?xml version="1.0" encoding="UTF-8"?>`

Note: The import fails if the encoding of the input file is in ANSI format. The input file must be in UTF-8 encoding.

- It must have one root element.
- Each element must consist of an open tag, `<tagName>`, a close tag, `</tagName>`, and the element content. The valid content of each element is defined in the DTD.
- Element tags are case sensitive.
- You can nest elements, but not overlap them. For example, this shows properly nested elements:

```
<tag1><tag2>theName</tag2></tag1>
```

This shows overlapping elements that result in an error:

```
<tag1><tag2>theName</tag1></tag2>
```

- If you do not form the import file correctly, the import fails. A message prints in the output file indicating the reason for the failure.

The import file may contain blank space, blank lines, and tabs, and so on between elements. Blank space is ignored unless it is part of a text string. For example, in some cases all tags and all children elements are shown on the same line, whereas in other cases they are shown on separate lines. Blank space at the beginning and the end of a field is ignored. Fields containing only blank space are considered empty.

XML version 1.0 is supported. Updates are made as necessary to support future versions of XML.

Note: NICE recommends you validate the XML file before uploading it.

◆ File Elements

Each import type has its own root element. For example, for agent import, the root element is **agentList**. You must nest all other elements within the root element. If the root element does not exist, the import fails.

If any elements exist outside the root element, the file is not considered well-formed and the import fails. A message is included in the output file indicating the reason for the failure.

◆ Element Data Types

The following standard XML data types are used:

```
<!ELEMENT elementName (childElement, childElement, ...)>
```

When you define an element with a list of children elements separated by commas, the children must appear in the same order in the import file. Otherwise, the element is not considered valid.

```
<!ELEMENT elementName (#PCDATA)>
```

Elements with this data type may contain any text string. The import program validates the text as necessary.

◆ Element Qualifiers

Standard XML element qualifiers determine whether each element is required and the number of times an element may exist:

- Elements marked with ? in the DTD are not required. If you include these elements, you can only specify them one time for each agent.
- Elements marked with * are not required. You can include more than one of these elements for each agent.
- Elements marked with + are required. You can include more than one of these elements for each agent, but at least one is required.
- Elements not marked with a qualifier are required and you can only specify them one time for each agent.

If an element does not meet these conditions, it is considered invalid.

◆ Element Attributes

Standard XML attribute rules are used. Attributes must be quoted. If an attribute is not contained in double quotes, the element is considered invalid.

If an attribute marked as **#REQUIRED** is missing from the associated element, the element is considered invalid.

◆ Comments

You can enter comments in the following format:

```
<!--comment text... -->
```

All text contained between `<!--` and `-->` is ignored by the import program.

4. Import Agent

The Import Agent feature allows you to import values that display in the *Agent Definition* view from an XML formatted source file. You can use Import Agent to add new agents or to change information for agents already defined in the WFM database.

Import Agent:

- reduces the data entry time required to define agents during the initial database build.
- reduces the data entry time required to add and change agents.
- supports adding and changing agents from a third party system, such as a human resource database.

This section includes this information for Import Agent:

- Import Agent file specifications for adding new agents or changing existing agents in the WFM database.
- Input file samples for adding new agent input files and changed agent input files.
- Command line parameters to run the Import Agent program.
- Import Agent processing and how new agent and changed agent records are processed.
- Import Agent DTD for the supported releases.

Note: Implementation details to automate the import of agent definition data are not included.

4.1 Agent Names and Passwords

4.1.1 Agent Names

You can use the single **name** element or you can import agent names into separate fields using the **firstName**, **lastName**, and **nameSuffix** elements. If you use the single name element to import names, the entire name is imported into the last name field.

Name element formats:

Single Name	Separate Fields
<pre><agentList> <newAgent> <name>last, first</name> </newAgent> </agentList></pre>	<pre><agentList> <newAgent> <lastName>last</lastName> <firstName>first</firstName> <nameSuffix >suffix</nameSuffix> </newAgent> </agentList></pre>

Single Name	Separate Fields
<pre> <agentList> <changedAgent> <name>last, first</name> </changedAgent> </agentList> </pre>	<pre> <agentList> <changedAgent> <lastName>last</lastName> <firstName>first</firstName> <nameSuffix >suffix</nameSuffix> </changedAgent> </agentList> </pre>

4.1.2 Agent Passwords

You cannot import passwords. However, NICE strongly recommends you set an email address for every agent. Also, when importing a new agent, you can specify the UUID. If you don't specify a UUID, WFM creates a UUID for the agent. Once the UUID is created, it cannot be changed.

4.2 Import Agent File Specifications

4.2.1 File Name and Location Specifications

To avoid potential problems with processing input/output files, it is recommended you limit the input/output file name to 30 characters or less.

You should put the input file in the specified directory. See [3.1 Import File Name and Location on page 14](#).

4.2.2 Time and Time Zones

Date and time elements referenced in Import Agent refer to time in the WFM server's time zone.

4.2.3 File Contents – General Specifications

◆ File Elements

The root element for import agent is **agentList**.

There are two types of elements you can include in the import file, within the **agentList** element:

- **newAgent** - a new agent is added.
- **changedAgent** - the import program searches for an existing agent in the WFM database that matches the specified agent ID, personal ID, or ACD/login combination. The match criterion used depends on the **-a** command line parameter. If a match is found, the changes specified in the **changedAgent** element are made. If no match is found, a message prints in the output file.

◆ General Data Requirements

In your XML file, **elements must be in the same order as listed in the corresponding DTD**. Although many are optional, you must put the present elements in the correct order.

Minimum data requirements for **newAgent** elements:

name, mu

Minimum data requirements for **changeAgent** elements:

id

4.2.4 Import Agent DTD

The Import Agent DTD contains the elements available to use in the import XML file to specify the information you want to import.

Note: Do not create a DTD file for the import; only use this as a template for building XML input files.

```
<!ELEMENT agentList (newAgent | changedAgent)* >

<!ELEMENT newAgent ((name | (lastName, firstName?, nameSuffix?)), UUID?, mu, acdList?,
    startID?,ssn?,agentStartDate?, externalID?,senDate?, senExt?, rank?, supervisor?,
    minTimeBetweenScheds?, minRestPerWeek?,minCDW?, maxCDW?, minCDO?, maxCDO?,availDateRange?,
    forceWork?, workStatus?, tourGroup?, rotationNum?, minMinutes?, maxMinutes?, useAvailability?,
    availListDateRange?, availabilityList?, skillSet?, agentData?, emailList?, vacationGroup?,
    biddingDate?, biddingExt?, accrualDate?, daysVacWeek?, vacDayLengthList?, recid?)>

<!ELEMENT changedAgent (id, (name | (lastName, firstName?, nameSuffix?))?, UUID?, mu?, acdList?,
    ssn?,externalID?, senDate?, senExt?, rank?, supervisor?, minTimeBetweenScheds?,
    minRestPerWeek?, minCDW?, maxCDW?, minCDO?, maxCDO?, availDateRange?, forceWork?, workStatus?,
    tourGroup?, rotationNum?, minMinutes?, maxMinutes?, useAvailability?, availListDateRange?,
    availabilityList?, skillSet?, agentData?, emailList?, biddingDate?, biddingExt?, accrualDate?,
    daysVacWeek?, vacDayLengthList?, recid?)>

<!ELEMENT id (#PCDATA)>
<!ELEMENT name (#PCDATA)>
<!ELEMENT lastName (#PCDATA)>
<!ELEMENT firstName (#PCDATA)>
<!ELEMENT nameSuffix (#PCDATA)>
<!ELEMENT UUID (#PCDATA)>
<!ELEMENT mu (#PCDATA)>
<!ELEMENT acdList (acdStartDate?, acd+)>
<!ELEMENT acdStartDate (day, month, year)>
<!ELEMENT acd (acdID, logon?)>
<!ELEMENT acdID (#PCDATA)>
<!ELEMENT logon (#PCDATA)>
<!ELEMENT startID (#PCDATA)>
<!ELEMENT ssn (#PCDATA)>
<!ELEMENT agentStartDate (day, month, year)>
<!ELEMENT externalID (#PCDATA)>
<!ELEMENT senDate (day, month, year)?>
<!ELEMENT senExt (#PCDATA)>
<!ELEMENT rank (#PCDATA)>
<!ELEMENT supervisor (#PCDATA)>
<!ELEMENT minTimeBetweenScheds (#PCDATA)>
<!ELEMENT minRestPerWeek (#PCDATA)>
<!ELEMENT minCDW (#PCDATA)>
<!ELEMENT maxCDW (#PCDATA)>
<!ELEMENT minCDO (#PCDATA)>
<!ELEMENT maxCDO (#PCDATA)>
<!ELEMENT availDateRange ((day, month, year, weeks) | (startDate, endDate) | default)?>
<!ELEMENT forceWork (#PCDATA)>
<!ELEMENT workStatus (#PCDATA)>
<!ELEMENT tourGroup (#PCDATA)>
<!ELEMENT rotationNum (#PCDATA)>
<!ELEMENT minMinutes (#PCDATA)>
<!ELEMENT maxMinutes (#PCDATA)>
<!ELEMENT useAvailability (#PCDATA)>
<!ELEMENT availListDateRange ((day, month, year, weeks) | (startDate, endDate) | default)>
<!ELEMENT availabilityList (availability+)>
<!ELEMENT availability (#PCDATA)>
<!ELEMENT skillSet (skillStartDate?, baseScheduleCode?, skill*)>
```

```

<!ELEMENT baseScheduleCode (#PCDATA)>
<!ELEMENT skillStartDate (day, month, year)>
<!ELEMENT skill (skillName, level?, rsv?)>
<!ELEMENT skillName (#PCDATA)>
<!ELEMENT level (#PCDATA)>
<!ELEMENT rsv (#PCDATA)>
<!ELEMENT agentData (group+)>
<!ELEMENT group (description, adgStartDate?, value?)>
<!ELEMENT description (#PCDATA)>
<!ELEMENT adgStartDate (day, month, year)?>
<!ELEMENT value (#PCDATA)>
<!ELEMENT emailList (email+)>
<!ELEMENT email (#PCDATA)>
<!ELEMENT vacationGroup (#PCDATA)>
<!ELEMENT biddingDate (day, month, year)?>
<!ELEMENT biddingExt (#PCDATA)>
<!ELEMENT accrualDate (day, month, year)?>
<!ELEMENT daysVacWeek (#PCDATA)>
<!ELEMENT vacDayLengthList (vacDayLength+)>
<!ELEMENT vacDayLength (#PCDATA)>
<!ELEMENT recid (#PCDATA)>
<!ELEMENT startDate (day, month, year)>
<!ELEMENT endDate (day, month, year)>
<!ELEMENT day (#PCDATA)>
<!ELEMENT month (#PCDATA)>
<!ELEMENT year (#PCDATA)>
<!ELEMENT weeks (#PCDATA)>
<!ELEMENT default EMPTY>

<!ATTLIST id acdID CDATA #IMPLIED>
<!ATTLIST logon logonType (AgentID | Extension | Email | ChatAlias) "AgentID">
<!ATTLIST availability day (Sun | Mon | Tue | Wed | Thu | Fri | Sat | All) "All">
<!ATTLIST availabilityList status (Approved | Rejected | Pending) "Approved">
<!ATTLIST email number CDATA #REQUIRED>
<!ATTLIST email type (Normal | Short) "Normal">
<!ATTLIST email language CDATA "English">
<!ATTLIST vacDayLength day (Sun | Mon | Tue | Wed | Thu | Fri | Sat | All) "All">

```

4.2.5 Import Agent File Contents – Record Specifications

These tables include all elements available, from the Import Agent DTD, to include in the Import Agent XML file. See [4.2.4 Import Agent DTD on page 19](#). The tables are separated based on specific agent information, definition, skills, agent data group, email, time off, and WFM databases. The tables consist of the element names, indented based on their level in the DTD, a description of the elements, and valid values for the elements. If you can edit an element in the WFM user interface, the **Valid Values** column references the WFM view.

Definition - This information from the DTD pertains to agent definition information.

Field	Description	Valid Values
Id<ID>	For changedAgent elements, this is the identifier used to find the agent to modify in NICE WFM. This may be a personal ID, a WFM agent ID, or an ACD logon ID depending on the -a command line parameter. Not used in newAgent elements.	S - 256 alphanumeric characters T - 1-999999999 A - Any printable ASCII characters except & and \. Maximum length is 64. For information on these parameters, see 4.4.1 Import Agent

Field	Description	Valid Values
		Command Line Parameters on page 36. If you use the A or S parameter, can be edited on view: <i>Agent Definition</i>
acdID (attribute of id)	If you use the -aA command line parameter, you must include this field to define the ACD for which the agent is assigned the logon ID specified in the id element above. There must be an ACD defined in NICE WFM with this ID. The agent assigned the specified logon for the specified ACD on the current date, based on the server time zone, is considered a match. This field is not required if you use the -aS , or -aT command line parameter.	1-2147483647 Can be edited on view: <i>Agent Definition</i>
name	Use one of the following: Option 1: <code><name>John Smith PhD</name></code> The name John Smith PhD is imported to the last name field. Option 2: <code><lastName>Smith</lastName></code> <code><firstName>John</firstName></code> <code><nameSuffix>PhD</nameSuffix></code> John is imported to the first name field. Smith is imported to the last name field. PhD is imported to the suffix field.	Any printable ASCII characters except and \. If you specify a name longer than 64 characters in one of the name elements, an error is generated. Note: If you include an ampersand (&), you must format it as: &amp; Can be edited on view: <i>Agent Definition</i>
UUID	The optional unified user ID for the agent. It is a unique ID used to synchronize the agent within the NICE products. The UUID must be unique for agents in the same customer ID. If a UUID is specified for either a changedAgent element or a newAgent element, and the specified ID is already assigned to another agent or system user in the same customer, the agent is not imported. Any letters in the UUID are saved in lower case.	Alphanumeric; maximum 128 characters.

Field	Description	Valid Values
mu	There must be an MU defined in NICE WFM with this ID. For newAgent elements, this is the ID of the MU to assign to the new agent. For changedAgent elements, if the mu element is present, the Move Agents process is invoked to change the agent's MU. Note: Moving an agent to a new MU always uses the current date.	1-999999999 Can be edited on view: <i>Agent Definition, Move Agent</i>
acdList	This element contains the ACDs to assign to the agent, as well as the associated logon ID and agent group. For changedAgent elements, these ACDs are not added to existing ACD assignments; they replace existing assignments, beginning on the acdStartDate. Note: The order you list the ACDs in the file determines the <i>priority</i> setting for each ACD, if you import more than one ACD.	This element can contain between 1 and 10 acd children. You can include one acdStartDate child, regardless of MU type. Can be edited on view: <i>Agent Definition, Move Agent</i>
acdStartDate	The beginning date for which the ACD assignments apply. If you do not specify an acdStartDate, the later of the current date and the earliest s_date for any of the agent's active MURoster records is used. Note: It is recommended to omit this field for newAgent elements; the importer defaults to the current date when you omit this field, which is the same functionality as the <i>Agent Definition</i> view.	Any valid date formed by day, month, and year children. This date must be on or after the agentStartDate. Can be edited on view: <i>Agent Definition, Move Agent</i>
day	The day of the acdStartDate .	1-31
month	The month of the acdStartDate .	1-12
year	The year of the acdStartDate .	Any year
acd	The ID of an ACD to assign to the agent, and optionally the agent's logon on the ACD.	acdID and logon children. Can be edited on view: <i>Agent Definition, Move Agent</i>
acdID	The ID of an ACD to assign to the agent. There must be an ACD defined in NICE	1-2147483647 Can be edited on view:

Field	Description	Valid Values
	WFM with this ID. This is the ACD ID, not port ID.	<i>Agent Definition, Move Agent</i>
logon	The agent's logon ID on the ACD specified in the same acd element. Agent logon IDs must be unique for the ACD, for all dates. If you do not specify an ID, a null value is used. If another agent is defined with the same logon on the same ACD, for a date on or after the specified start date, the agent is not imported. Note: There may be existing agents in the database with a null logon value; if the imported agent also has a null logon value, the agent is imported.	Blank or any printable ASCII characters except & and \. Maximum length is: ▪ 100 if ACD logonType = AgentID or Extension ▪ 200 if ACD logonType = Email or ChatAlias Can be edited on view: <i>Agent Definition, Move Agent</i>
logonType (attribute of logon)	The type of logon used for synchronizing agents between NICE products. This optional ACD login type defaults to AgentID, if not specified for an ACD login.	AgentID Extension Email ChatAlias Can be edited on view: <i>Agent Definition, Move Agent</i>
startID	If you use the -i command line parameter, you must use the specified startID in the newAgent element as the agent's WFM ID when defining a new agent using the import. If you do not specify a startID, or the specified startID is not available, the new agent is not defined and an error message is printed in the output file indicating the startID was invalid or not available. If you do not use the -i parameter and you include a startID element, the new agent's WFM ID is the next available ID starting with the given startID. If you do not use this parameter and you do not include a startID element, the new agent's WFM ID is the next available ID. Not used in changedAgent elements.	1-999999999
ssn	The agent's personal ID.	256 alphanumeric characters. Can be edited on view: <i>Agent Definition, Agent Profile</i>

Field	Description	Valid Values
agentStartDate	The agent's start date with your company. Not used in changedAgent elements. Note: It is recommended you omit this field for newAgent elements; the importer defaults to the current date when you omit this field, which is the same functionality as the <i>Agent Definition</i> view).	Valid date, formed using day, month, and year children. This date cannot be later than the current date. This date cannot be earlier than the first date for which schedules exist in the MU. If any dbAgentMap / MURoster records exist with this agent ID as this new agent's ID, the earliest valid start date is either the day after the latest s_date or e_date in those records or the first date for which schedules exist in the MU, whichever date is latest. Only the dbAgentMap / MURoster records whose recStatus / C_Act fields are active are considered. Note: These are the same validations used in the <i>Agent Definition</i> view. Can be edited on view: <i>Agent Definition</i>
day	The day of the agentStartDate .	1-31
month	The month of the agentStartDate .	1-12
year	The year of the agentStartDate .	Any year
externalID	The ID the agent uses for logging into the Agent WebStation, if the WebStation is configured to use external agent IDs. The external ID and agent ID must be unique within a customer.	Blank or any printable ASCII characters except & and \. Maximum length is 256. Can be edited on view: <i>Agent Definition, Agent Profile</i>
senDate	The agent's seniority date.	Blank or valid date, formed using day, month, and year children. Any date. Can be edited on view: <i>Agent Definition, Agent Profile</i>
day	The day of the agent's senDate .	1-31
month	The month of the agent's senDate .	1-12
year	The year of the agent's senDate .	Any year
senExt	The agent's seniority extension.	Blank or 0-9999 Can be edited on view:

Field	Description	Valid Values
		<i>Agent Definition, Agent Profile</i>
rank	The agent's rank.	Blank or 0-9999 Can be edited on view: <i>Agent Definition, Agent Profile</i>
supervisor	Values provided in this element are not stored in the database.	Not applicable
minCDO	The minimum number of consecutive days off for the employee.	Blank or 1-14 Can be edited on view: <i>Agent Definition, Agent Availability</i>
maxCDO	The maximum number of consecutive days off for the employee.	Blank or 1-14 Can be edited on view: <i>Agent Definition, Agent Availability</i>
minCDW	The minimum number of consecutive days the employee may work.	Blank or 1-14 Can be edited on view: <i>Agent Definition, Agent Availability</i>
maxCDW	The maximum number of consecutive days the employee may work.	Blank or 1-14 Can be edited on view: <i>Agent Definition, Agent Profile, Agent Availability</i>
minTimeBetweenScheds	The minimum number of unscheduled minutes required between the agent's schedules. Value must be in increments of the scheduling period length.	Blank or 0-1440 Can be edited on view: <i>Agent Definition, Agent Profile, Agent Availability</i>
minRestPerWeek	The minimum number of unscheduled minutes required in a continuous block, at least once per week. Value must be in increments of the scheduling period length.	Blank or 0-10080 Can be edited on view: <i>Agent Definition, Agent Availability</i>
availDateRange	The beginning date and number of weeks for which the availability data applies. Availability data includes minCDO, maxCDO, minCDW, maxCDW, minTimeBetweenScheds, minRestPerWeek, forceWork, workStatus, tourGroup, rotationNum, minMinutes, maxMinutes, useAvailability, and availabilityList elements. This must be one of these: ▪ a date plus a number of weeks ▪ a start date and an end date ▪ an empty <default> element	Blank or any valid date, formed using day, month, and year children and one weeks child or startDate and endDate elements formed using day, month, and year children or A <default> element Can be edited on view: <i>Agent Definition, Agent Availability</i>

Field	Description	Valid Values
	The startDate and endDate range can be any length. If you include this element with no value or if it contains a <default/> element, data is imported for the agent's Default record.	
day	The day of the availDateRange .	1-31 Date is not required to fall on the first day of week defined for the agent's scheduling entity.
month	The month of the availDateRange .	1-12
year	The year of the availDateRange .	Any year
weeks	The number of weeks to apply the availability information.	1-999 Can be edited on view: <i>Agent Definition, Agent Availability</i>
startDate	The date to start using the scheduling parameters.	Can be edited on view: <i>Agent Definition, Agent Availability</i>
day	The day of the startDate .	1-31
month	The month of the startDate .	1-12
year	The year of the startDate .	Any year
endDate	The date to stop using the scheduling parameters.	Can be edited on view: <i>Agent Definition, Agent Availability</i>
day	The day of the endDate .	1-31
month	The month of the endDate .	1-12
year	The year of the endDate .	Any year
default	This indicates to update the Default record.	Empty element Can be edited on view: <i>Agent Definition, Agent Availability</i>
forceWork	Whether to honor the agent's minimum work rules.	Blank, Y , or N Not case sensitive. Can be edited on view: <i>Agent Definition, Agent Availability</i>
workStatus	Whether the agent works full-time or part-time.	Blank, F (Full), or P (Part) Not case sensitive. Can be edited on view: <i>Agent Definition, Agent Availability</i>
tourGroup	The ID of the agent's weekly rule. There must be a weekly rule defined in NICE WFM with this ID.	Blank or 1-9999 Can be edited on view: <i>Agent Definition, Agent Availability</i>
rotationNum	The agent's weekly rule rotation number. This must be a valid rotation number for the weekly rule.	Blank or 1-99 Can be edited on view: <i>Agent Definition, Agent Availability</i>

Field	Description	Valid Values
	If this element is blank: ▪ it is ignored for normal weekly rule ▪ it causes an error if the weekly rule is rotating	
minMinutes	The minimum amount of time the agent must work in a week, in minutes.	Blank or 0-10080 Can be edited on view: <i>Agent Definition, Agent Availability</i>
maxMinutes	The maximum amount of time the agent may work in a week, in minutes.	Blank or 0-10080 Can be edited on view: <i>Agent Definition, Agent Availability</i>
useAvailability	Values provided in this element are not stored in the database. Agent availability is always used.	Not applicable Can be edited on view: <i>Agent Definition, Agent Availability</i>
availListDateRange	Use this optional element to store the availability data for a date range that is different than the other scheduling parameters. If you do not provide an availListDateRange element, the availDateRange is used for storing availability data. If you do not provide an availDateRange, the Default record for availability is updated. This must be one of these: ▪ a date plus a number of weeks ▪ a start date and an end date ▪ an empty <default> element The startDate and endDate range can be any length. If you include this element without a value or if this element contains a <default/> element, data is imported for the agent's Default record.	Any valid date, formed using day, month, and year children and one weeks child or startDate and endDate elements formed using day, month, and year children or a <default> element Can be edited on view: <i>Agent Definition, Agent Availability</i>
day	The day of the availListDateRange.	1-31 Date is not required to fall on the first day of week defined for the agent's scheduling entity.
month	The month of the availListDateRange .	1-12
year	The year of the availListDateRange .	Any year
weeks	The number of weeks to apply the availability information.	1-999 Can be edited on view:

Field	Description	Valid Values
		<i>Agent Definition, Agent Availability</i>
startDate	The date to start using the scheduling parameters.	Can be edited on view: <i>Agent Definition, Agent Availability</i>
day	The day of the startDate .	1-31
month	The month of the startDate .	1-12
year	The year of the startDate .	Any year
endDate	The date the scheduling parameters will cease to be used.	Can be edited on view: <i>Agent Definition, Agent Availability</i>
day	The day of the endDate .	1-31
month	The month of the endDate .	1-12
year	The year of the endDate .	Any year
default	Update the Default record.	Empty element Can be edited on view: <i>Agent Definition, Agent Availability</i>
availabilityList	This element contains availability children. You can specify availability data for as many days of the week as desired.	availability children Can be edited on view: <i>Agent Definition, Agent Availability</i>
status (attribute of availabilityList)	The status of the Availability record.	Approved Rejected Pending Can be edited on view: <i>Agent Definition, Agent Availability</i>
availability	This string contains the agent's availability for each period of the specified day of the week. If the day attribute is All , this availability data is stored for each day of the week.	Blank, a 96-character string in 15-minute increments, or a 288-character string in 5-minute increments. Each character must be either A or N . Not case sensitive. Can be edited on view: <i>Agent Definition, Agent Availability</i>
day (attribute of availability)	The day of the week for which the availability data applies.	Sun, Mon, Tue, Wed, Thu, Fri, Sat, or All Case is sensitive. Can be edited on view: <i>Agent Definition, Agent Availability</i>

Skill - This information from the DTD pertains to an agent's skill assignment. You can create a new skill assignment beginning with the date specified below and without an end date, but you cannot modify an existing date-specific skill assignment.

Field	Description	Valid Values
skillSet	The agent's skill assignments. For changedAgent elements, these skills are not added to existing skill assignments; they replace existing skill assignments, beginning on the specified skillStartDate. If you include this element without a skill child element, the agent's existing skills are deleted.	One optional skillStartDate child, one optional baseScheduleCode child. Between 0 and 50 skill children. Can be edited on view: <i>Agent Definition, Agent Profile</i>
skillStartDate	The beginning date for which the skill assignments apply. This element is required for changedAgent elements that contain a skillSet element. Note: It is recommended you omit this field for newAgent elements. When omitted the importer defaults to the current date, which is the same functionality as the <i>Agent Definition</i> view.	Any valid date formed by day, month, and year children This date must be on or after the agentStartDate. Can be edited on view: <i>Agent Definition, Agent Profile</i>
day	The day of the skillStartDate .	1-31
month	The month of the skillStartDate .	1-12
year	The year of the skillStartDate .	Any year
baseScheduleCode	The activity code used as the agent's base code when generating schedules. If you do not provide this element, the Open (blue icon) activity code is used. Since base schedule codes are stored with agent availability data, the baseScheduleCode is stored with the date range specified in the availDateRange element, rather than the skillStartDate element. If you do not include the availDateRange element, the baseScheduleCode is stored in the default Availability record.	Blank or any printable ASCII characters except & and \ Case is sensitive. Value must match a defined activity code description. Maximum length is 64. Can be edited on view: <i>Agent Definition, Agent Profile</i>
skill	This element contains information about a skill assigned to the agent.	One skillName, one optional level child, and one optional rsv child. Can be edited on view: <i>Agent Definition, Agent Profile</i>
skillName	The name of the skill assigned to the agent. There must be a skill defined in NICE WFM with this name.	Any printable ASCII characters except & and \ Case is sensitive. Maximum length is 64. Can be edited on view: <i>Agent Definition, Agent Profile</i>

Field	Description	Valid Values
level	The level assigned to the agent for the associated skill.	Blank or 1-50 Can be edited on view: <i>Agent Definition, Agent Profile</i>
rsv	Indicates if the agent is a reserve agent for the associated skill.	Blank, Y , or N Case is sensitive. Can be edited on view: <i>Agent Definition, Agent Profile</i>

Agent Data Group - This information from the DTD pertains to an agent's agent data group assignments:

Field	Description	Valid Values
agentData	This element contains all the agent data group information for the agent.	group children Can be edited on view: <i>Agent Data</i>
group	This child element contains the information for one agent data group.	description , adgStartDate , and value children Can be edited on view: <i>Agent Data</i>
description	The description of the agent data group for which the agent is assigned a value . There must be an agent data group defined with this description for the MU specified in the CustID . Note: If you include an ampersand (&), you must format it as &amp; ;	Any printable ASCII characters except and \ Case is sensitive. Maximum length is 64. Can be edited on view: <i>Agent Data</i>
adgStartDate	The start date for the value assigned to the agent for the agent data group. - If the adgStartDate is blank or you omit the element, the current date is used. - If the adgStartDate is invalid, the record is not imported. - If the agent has any existing date ranges for the agent data group: - Date ranges whose start dates fall on or after the specified start date are deleted. - If a date range exists whose start date falls before the specified start date and whose end date is either null, the same as, or later than the specified start date, the end date of this record is changed to the day before the specified start date. - The agent data value is stored with the specified start date and the end date is null.	Date, formed using day , month , and year children. Can be edited on view: <i>Agent Data</i>

Field	Description	Valid Values
day	The day of the agent's adgStartDate .	1-31
month	The month of the agent's adgStartDate .	1-12
year	The year of the agent's adgStartDate .	Any year
value	The value assigned to the agent for the associated agent data group. - For changedAgent elements, this value replaces any previously assigned value. - This value must be valid for the specified agent data group. In other words, it must be a date for date-type groups [format: YYYYMMDD], a pre-defined list value for a list-type group, and within the min/max constraints, and so on. - If the agent data group's data type is system user, this value should be a user's login name. - No value is required for changedAgent elements, whereas you must specify a value for newAgent elements if you include the agentData element. Note: If you include an ampersand (&), you must format it as &amp;;	Blank or any printable ASCII characters except and \ Case is sensitive. Maximum length is 64. Can be edited on view: <i>Agent Data</i>

Email - This information from the DTD pertains to an agent's email address assignments:

Field	Description	Valid Values
emailList	One or more email addresses for the agent. Only one email address is imported. If you include more than one email address in the input file, only the first address is imported. All other addresses are ignored.	email children
email	This child element contains one email address.	Any printable ASCII characters except & and \. Maximum length is 64. Can be edited on view: <i>Agent Definition, Agent Profile</i>
number (attribute of email)	This element is required. Only one email address is imported. If you include more than one email address in the input file, only the first address is imported. All other addresses are ignored.	1
type (attribute of email)	This element is not required. If you include it, it is ignored. All email is sent using the Short type.	Short

Field	Description	Valid Values
language (attribute of email)	This element is not required. The language, the language and country combination, or the 2-5 character <i>locale</i> used for the email address. The setting in this field determines the language in which email is sent to the address, as well as the date and time formats used. A record must exist for the customer in the R_TVLOCALE table whose C_LOCALE field matches the string specified in the language attribute. You can specify the language and country in any language supported by the WFM workstation. A locale is the 2-5 character code that indicates language and country, as defined by ISO-639 and ISO-3166. You can find a full list of language codes at http://www.loc.gov/standards/iso639-2/php/English_list.php. You can find a full list of country codes at http://www.iso.org/iso/country_codes.	Any printable ASCII characters except & and \ Not case sensitive. Maximum length is 60.

Time Off - This information from the DTD pertains to an agent's time off data:

Field	Description	Valid Values
vacationGroup	The ID of the time off group to assign to the new agent. There must be a time off group defined in NICE WFM with this ID. Not used in changedAgent elements.	1-2147483647 Can be edited on view: <i>Agent Definition, Agent Profile</i>
biddingDate	The agent's alternate seniority date.	Blank or any valid date formed by day, month, and year children. Can be edited on view: <i>Agent Definition, Agent Profile</i>
day	The day of the agent's biddingDate .	1-31
month	The month of the agent's biddingDate .	1-12
year	The year of the agent's biddingDate .	Any year
biddingExt	The agent's alternate seniority extension.	Blank or 0-9999. Can be edited on view: <i>Agent Definition, Agent Profile</i>
accrualDate	The agent's vacation accrual date.	Blank or any valid date formed by day, month, and year children. Can be edited on view: <i>Agent Definition, Agent Profile</i>

Field	Description	Valid Values
day	The day of the agent's accrualDate .	1-31
month	The month of the agent's accrualDate .	1-12
year	The year of the agent's accrualDate .	Any year
daysVacWeek	The number of vacation days that comprise a week of vacation for the agent.	Blank or 0-7 Can be edited on view: <i>Agent Definition, Agent Profile</i>
vacDayLengthList	This element contains the length of a vacation day for the days of the week. You can specify this data for as many days of the week as necessary.	vacDayLength children Can be edited on screen: <i>Agent Definition, Agent Profile</i>
vacDayLength	The number of minutes of vacation the agent gets when taking a full day of vacation on the specified day of the week.	Blank or 0-1440 Can be edited on view: <i>Agent Definition, Agent Profile</i>
day (attribute of vacDayLength)	The day of the week for which the vacation day length is specified.	Sun, Mon, Tue, Wed, Thu, Fri, Sat, or All. Case is sensitive. Can be edited on view: <i>Agent Definition, Agent Profile</i>

WFM Databases - This information from the DTD does not pertain to any data in the WFM databases:

Field	Description	Valid Values
Recid	Provided for you to assign a unique record ID to each add or change element in the XML file. You are responsible to ensure the ID is unique or otherwise interpretable by your external system. This element is not recorded in any NICE WFM table / field. See 4.6.2 If you include the -o command line parameter on page 43 for information on how recid is used.	Blank or any value

Note: You can import this vacation planner information using Import Vacation Summary:

- Earned
- Credited
- Carryover from last year
- Debited
- Carryover to next year

The **Planned** value is system populated from agent bidding and future activities; the **Taken** value is system populated from agent schedules; **Remaining** is system calculated.

4.3 Import Agent XML Input File Example

Note: Data in all fields must match the format used in the RCP.

Example for new agents:

Almost all the fields are specified for this **newAgent** element; however, it is not necessary to include all fields.

[illegible]

[illegible]

Example for changed agents:

This example assumes the **-a** command line parameter is **S** which indicates personal ID is used to identify agents. Only the fields to modify are included in these example elements.

```
<?xml version="1.0" encoding="UTF-8" ?>
<agentList>

    <changedAgent>
        <id>123456789</id>
        <acdList>
            <acdStartDate>
                <day>1</day><month>2</month><year>2021</year>
            </acdStartDate>
            <acd><acdID>25</acdID><logon>456</logon></acd>
            <acd><acdID>999</acdID></acd>
        </acdList>
    </changedAgent>

```

```

        </acdList>
        <skillSet>
            <baseScheduleCode>Phones</baseScheduleCode>
            <skillStartDate>
                <day>1</day><month>2</month><year>2021</year>
            </skillStartDate>
            <skill><skillName>Sales</skillName><level>1</level></skill>
            <skill><skillName>Service</skillName><level>2</level></skill>
        </skillSet>
        <agentData>
            <group><description>Team Ldr</description><value>Mike Shaw</value></group>
        </agentData>
        <emailList>
            < email number="1">name@company.com</email>
        </emailList>
    </changedAgent>

    <changedAgent>
        <id>987654321</id>
        <lastName>Barni</lastName>
        <firstName>Meghan</firstName>
    </changedAgent>
</agentList>

```

4.4 Import Agent Command Line

For distributed systems, including hosted/cloud environments, copy the import file onto the EFT site before executing the import agent command. A process automatically moves the file from the EFT site to the server. It can take SmartSync 10 minutes or more to process files depending on the size of the files and the timing of tasks to move files.

You can execute the program manually or automatically on a pre-determined schedule.

Note: The Import Agent input file is deleted after a successful import, unless you run the `importAgent` command using the `-k` command line parameter. If the file cannot be deleted, this message appears in the output file: `Could not delete input file.`

4.4.1 Import Agent Command Line Parameters

Import Agent is invoked using the `importAgent` command and these parameters:

Parameter	Description
-c<custID> (required)	The ID of the customer partition for which to import agents. 1 - 32757
-f<path/file.name> (recommended)	Use this parameter to specify the path and name of the file to import. The recommended path and file name is /totalview/ftp/imports/customer<custID>/agents/new/agents.1.xml and /totalview/ftp/imports/customer<custID>/agents/changes/chagents.xml If you do not provide a file parameter on the command line, the default is /totalview/ftp/imports/agents/agents.<n>.xml where <n> is the customer number.
-a<ID type> (optional)	The criterion used to match agents in the XML file with agents in the WFM database. Possible values, which are not case sensitive, are: T or t - Use the WFM agent ID to identify agents. Values specified in id elements must be WFM agent IDs, see Id on page 20.

	▪ S or s - Use the personal ID to identify agents. Values specified in id elements must be unique personal identification numbers, see Id on page 20. ▪ A or a - Use the ACD login ID to identify agents. Values specified in id elements must include an acd attribute and the id element contains the associated login ID, see Id on page 20. For example, to identify the agent whose login ID on ACD 25 is 1234, the id element is: <pre><id acdID="25">1234</id></pre> Note: If ACD logon is used to identify agents, the agent who is assigned the specified logon for the specified ACD on the current date, based on the server time zone, is considered a match. If you don't include this parameter in the command line, the program uses -aT to identify agents. This parameter is not used for importing new agents; it is only used for modifying existing agents.
-w (optional)	The import is done in what-if mode. This means the import file is processed without making changes in the WFM database. The output file is written regardless of this parameter setting. If you omit this parameter, updates are made to the database.
-k (optional)	This parameter indicates to keep the input file for the agent import after a successful import. If you do not include this parameter, the input file is deleted after a successful import.
-i<force start ID> (optional)	• If you use this parameter, the specified startID in the newAgent element is used as the agent's WFM ID when defining a new agent using the import. If you don't specify a startID, or if the specified startID is not available, the new agent is not defined and an error message is printed in the output file indicating the startID was invalid or not available. • If you don't use this parameter and you include a startID element, the new agent's WFM ID is the next available ID starting with the given startID. • If you don't use this parameter and do not include a startID element, the new agent's WFM ID is the next available ID.
-o (optional)	This parameter indicates to create an .OUT file that contains every element processed. See 4.6 Import Agent Output File on page 41. Regardless of whether you include the -o parameter, if there are any errors, the Import Agent process automatically creates an .ERR file. Note: If you use the GDPR Compliance Ready feature, do not use the -o parameter.

4.4.2 Import Agent Command Line Examples

Refer to the *Administrator Guide - WFM* for details about running commands from the *Custom Activities* view.

Import Agent Test Mode – To turn on test mode, which checks the import for errors without making any changes to the WFM system, add **-w** to the command line:

Linux:

```
importAgent -c<custID> -  
f/totalview/ftp/imports/customer<custID>/agents/new/agents.1.xml -i -w >  
/totalview/ftp/imports/customer<custID>/agents/new/run.log
```

Windows:

```
importAgent -c<custID> -  
f/totalview/ftp/imports/customer<custID>/agents/new/agents.1.xml -i -w >  
/totalview/ftp/imports/customer<custID>/agents/new/run.log 2>&1
```

Import Agent with Specified ID**Linux:**

```
importAgent -c<custID> -  
f/totalview/ftp/imports/customer<custID>/agents/new/agents.1.xml -i >  
/totalview/ftp/imports/customer<custID>/agents/new/run.log
```

Windows:

```
importAgent -c<custID> -  
f/totalview/ftp/imports/customer<custID>/agents/new/agents.1.xml -i >  
/totalview/ftp/imports/customer<custID>/agents/new/run.log 2>&1
```

Import Agent without ID**Linux:**

```
importAgent -c<custID> -  
f/totalview/ftp/imports/customer<custID>/agents/new/agents.1.xml >  
/totalview/ftp/imports/customer<custID>/agents/new/run.log
```

Windows:

```
importAgent -c<custID> -  
f/totalview/ftp/imports/customer<custID>/agents/new/agents.1.xml >  
/totalview/ftp/imports/customer<custID>/agents/new/run.log 2>&1
```

Change Agent**Linux:**

```
importAgent -c<custID> -  
f/totalview/ftp/imports/customer<custID>/agents/changes/chagents.xml -aT >  
/totalview/ftp/imports/customer<custID>/agents/changes/run.log
```

Windows:

```
importAgent -c<custID> -  
f/totalview/ftp/imports/customer<custID>/agents/changes/chagents.xml -aT >  
/totalview/ftp/imports/customer<custID>/agents/changes/run.log 2>&1
```

4.5 Import Agent Processing

If you included the what-if command line parameter, **-w**, the import does not make any changes to the WFM database.

If an element contains a *blank* value or if you omit an element which correlates to a required database field in a default record, the existing value in the WFM database which corresponds to the element is cleared.

Processing immediately fails, no new agents are imported no existing agents are changed, and an error message prints in the .ERR file indicating the reason for the failure in these circumstances:

- the Import Agent feature is not enabled for the specified customer
- you omitted the customer ID command line parameter, **-c**, or the specified customer ID is invalid. The message `The customer ID is invalid.` prints.

The customer ID is considered invalid if either of these conditions is true:

- you did not specify a valid integer
- there is no customer in NICE WFM with the id specified
- the file specified with the **-f** command line parameter does not exist
- you included the agent identifier command line parameter, **-a**, but used something other than **S**, **s**, **A**, **a**, **T** or **t**

4.5.1 newAgent Elements

For **newAgent** elements, if you specified an invalid value in any field, the new agent is not added to the database. A message is included in the output file indicating the reason for the failure, along with the entire **newAgent** element. The reason for failing, instead of adding the agent using default values for the invalid fields, allows you to correct the problem and easily re-import the element without having to create another **changedAgent** element to correct the problem.

Importing the same file more than once creates duplicate agent records in the WFM database unless ACD logons are included in the import or you use the **-i** command-line parameter. ACD logons are the only values in the import that must be unique.

If the **acdStartDate** is not equal to the **agentStartDate**, or the current date if you omit **agentStartDate**, the record is rejected and an error message is provided to indicate the dates must be equal.

4.5.2 changedAgent Elements

For each **changedAgent** element, the import program searches in the WFM database for an existing agent that matches the specified **id** child element. The contents of the **id** depends on the **-a** command line parameter:

- **T** - Values specified in **id** elements must be WFM agent IDs.
- **S** - Values specified in **id** elements must be personal IDs.
- **A** - Each **id** element must include an **acd** attribute and the **id** element contains the associated login ID.

If a match is found, the changes specified in the **changedAgent** elements are made.

If no match is found, a message is printed in the output file, along with the entire **changedAgent** element, and no change is made.

If you use personal ID as the identifier and you specified a value in the **ssn** element, the program matches the agent with the personal ID specified in the **id** element and changes the agent's personal ID to the value specified in the **ssn** element. This process is not recommended.

If you use ACD logon to identify agents, the agent who is assigned the specified logon for the specified ACD on the current date, based on the server time zone, is considered a match. **Note:** If you specified a new logon or new ACD assignments for the agent, the program matches the agent with the ACD logon specified in the **id** element and changes the agent's ACD/logon assignments as specified in the **acdList** element. This process is not recommended.

For **changedAgent** elements, if an element contains a child that is not allowed or contains an invalid value in any field, the entire element is considered invalid and the **changedAgent** element is not imported. A message is included in the output file, along with the entire element.

◆ Changing an agent's MU

If the **changedAgent** element contains an **mu** child element changing the agent's MU assignment, the system changes the agent's MU assignment before making any other changes for the agent. The date of move is the current date.

*If no MU is defined in NICE WFM with the ID specified in the **mu** child element, the entire element is considered invalid and the **changedAgent** element is not imported. A message is included in the output file, along with the entire element.*

4.5.3 Import Agent Impacts on Forecast Allocations and Schedules

Importing agent information updates the *Agent Definition*, *Agent Profile*, and *Agent Availability* views, and so on in the WFM database. The import process does not invoke forecast generations, schedule generations, or simulations. Each time an Import Agent process completes, whether it is for **newAgent** or **changedAgent** records, you need to determine which processes need to be invoked using the NICE WFM application to update forecasts allocations and schedules to show the effects of the database updates. Depending on the type of data imported, you need to determine the processing to complete in NICE WFM. There are four processes to consider:

- Regeneration of forecast allocations
- Regeneration of schedules
- Simulation

If you do not perform one or more of these processes, you may see unexpected results in scheduled open values, contact and/or requirement allocations, multi-skill efficiency estimates, and so on.

At a minimum, you should manually invoke a Simulate Schedules process if schedules exist for the date range where **newAgent** or **changedAgent** records updated these data items:

- **acdID**

- **skillName**
- **level**
- **rsv**

You may optionally invoke a Forecast Allocation or Schedule Generation process, or manually reallocate contacts and/or requirements using the view if a forecast allocation exists for the date range where **newAgent** or **changedAgent** records updated these data items:

- **mu**
- **acdID**
- **skillName**
- **level**
- **rsv**

4.6 Import Agent Output Files

If any errors occur, the Import Agent process creates an error file in the same directory as the import file. It has the same name as the import file with the **.ERR** extension.

If you include the **-o** command line parameter, an output file is created in the same directory as the import file. It has the same name as the import file with the **.OUT** extension.

Both **.ERR** and **.OUT** files include:

- Comment lines, before the `<?xml version>` line, with this information:
 - the customer ID for which agents are imported
 - the path and name of the import file
 - the time the import began
 - the time the import completed

Example:

```
<!-- Customer ID: 2 -->
<!-- Import file:
/totalview/ftp/imports/customer<custID>/agents/new/agents.1.xml -->
<!-- Start time: 3/5/22 2:23 PM -->
<!-- End time: 3/5/22 2:23 PM -->

<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
...
```

- The **agentList** element from the input file with a successful import.

Example:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<agentList xmlns:ns2="com.iex.tv.smartsync.ws.agentimport">
  <newAgent operation="N">
    <status>Success</status>
    ...
  </newAgent>
</agentList>
```

- Errors that occur when processing agents appear in the output files with this information:
 - a. the **status** element with the text: Failure
 - b. the **messages** element with an error message indicating the reason for the failure
 - c. error codes are not included

Example:

```
<!-- Customer ID: 2 -->
<!-- Import file:
/totalview/ftp/imports/customer<custID>/agents/new/agents.1.xml -->
<!-- Start time: 10/16/21 12:45 PM -->
<!-- End time: 10/16/21 12:45 PM -->

<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<agentList xmlns:ns2="com.iex.tv.smartsync.ws.agentimport">
  <newAgent operation="N">
    <status>Failure</status>
    <messages>MU not found for ID: 1210000000.</messages>
    <messages>Agent start date invalid</messages>
    ...
  </newAgent>
</agentList>
```

4.6.1 The .ERR File

If any errors occur, the Import Agent process creates an error file in the same directory as the import file. It has the same name as the import file with the **.ERR** extension.

If the command line contains an invalid customer ID or if the import file does not exist, an error message prints in place of the comment lines at the top of the **.ERR** file and processing ends.

If an error occurs importing a **newAgent** element or a **changedAgent** element, a comment line or a **messages** element is included to indicate the cause of the error, followed by the entire element.

At the end of the error file, a summary of the imported **newAgent** and **changedAgent** elements displays with these data items:

- the number of elements processed
- the number of errors that occurred
- the number of elements imported
- the number of elements not imported

The error file is formatted as an XML file so you can re-import the elements that were in error, after correcting the errors.

Example .ERR File

```
<!-- Customer ID: 12 -->
<!-- Import file: /totalview/ftp/imports/customer<custID>/agents/new/agents.1.xml -
-->
<!-- Start time: 10/16/21 12:45 PM -->
<!-- End time: 10/16/21 12:45 PM -->

<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<agentList xmlns:ns2="com.iex.tv.smartsync.ws.agentimport">
  <changedAgent operation="C">
    <status>Failure</status>
    <messages>invalid character in agent name</messages>
    <id>12345</id>
    <name>John&th&n W&shington</name>
  </changedAgent>
  ...
</agentList>

<!-- summary data... -->

<!-- newAgent elements: -->
  <!-- 112 newAgent elements were processed -->
  <!-- 25 errors were found -->
  <!-- 105 newAgent elements were successfully imported -->
  <!-- 7 newAgent elements were not imported -->

<!-- changedAgent elements: -->
  <!-- 2 changedAgent elements were processed -->
  <!-- 0 errors were found -->
  <!-- 2 changedAgent elements were successfully imported -->
  <!-- 0 changedAgent elements were not imported -->
```

4.6.2 If you include the -o command line parameter

If you include the **-o** command line parameter, an output file is created in the same directory as the import file. It has the same name as the import file with the **.OUT** extension.

The output file contains a single **agentList** element with a record for each **newAgent** and **changedAgent** element. Each **newAgent** and **changedAgent** element contains these children:

recid - This is the record ID specified for the **newAgent** or **changedAgent** element. If you did not specify a record ID, this field is blank.

status - If the agent imported successfully, this element contains the text `Success`; otherwise the element contains the text `Failure`.

messages - This element is only included if the status is `Failure`. It contains an error message indicating the reason for the failure. Multiple message elements may exist if multiple errors occurred in the same **newAgent** or **changedAgent** element.

No summary comments are included in the **.OUT** file.

If you use the GDPR Compliance Ready feature, do not use the **-o** parameter.

Example .OUT File

```
<!-- Customer ID: 2 -->
<!-- Import file: /totalview/ftp/imports/customer<custID>/agents/new/agents.1.xml -->
<!-- Start time: 3/5/21 2:23 PM -->
<!-- End time: 3/5/21 2:23 PM -->
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>

<agentList xmlns:ns2="com.iex.tv.smartsync.ws.agentimport">
  <newAgent operation="N">
    <recid>10231</recid>
    <status>Failure</status>
    <messages>Agent seniority date not valid</messages>
    ...
  </newAgent>
  <changedAgent operation="C">
    <recid>10502</recid>
    <status>Failure</status>
    <messages>Can't find Agent with matching TotalView ID</messages>
    ...
  </changedAgent>
  <changedAgent operation="C">
    <recid>10502</recid>
    <status>Failure</status>
    <messages>Agent Name contains invalid characters or is too long</messages>
    <messages>Agent ACD not found in system</messages>
    ...
  </changedAgent>
  <newAgent operation="N">
    <recid>32200</recid>
    <status>Success</status>
  </newAgent>
</agentList>
```

Note: If you rerun the contents of the **.OUT** file, you must remove the **xmlns** attribute from the **agentList** element and the **operation** attribute from the **newAgent** or **changedAgent** elements.

5. Import Forecast

Import Forecast allows you to import contact forecast data into NICE WFM for one or more CTs.

- This feature is useful for workforce management groups that cut and paste spreadsheet data into the NICE WFM *Adjust Forecast* view.
- You can import daily or periodic values.
- Daily values are allocated to periods within the date based on the existing forecast distribution.
- You can import multiple CTs in the same data file.

For details on Digital Channel Management (DCM) and how the requirements calculator and simulation use DCM data, see **Digital Channel Management** in the *NICE WFM Training Manual for R8.x*.

This section covers these main topics for Import Forecast:

- Import Forecast file specifications for importing contact forecast data into the WFM database
- Input file sample for importing contact forecast data
- Command line parameters to run the Import Forecast program
- Import Forecast processing, including information on the **dailyData** and **periodData** elements, output file, error file, and lock file
- Import Forecast DTD
- MU requirements conversion utility

Note: The implementation details to automate the import of contact forecast data are not included.

5.1 Customer Responsibilities

You can provide the number of contacts, the AHT, or both. If you do not supply the contact or AHT data, the forecast data produced by the NICE WFM forecast generator is retained.

An error file is produced that includes the details of any record that could not be processed, along with the reason why it could not be processed. The program also reports the number of records successfully processed.

5.2 Import Forecast File Specifications

5.2.1 File Name and Location Specifications

To avoid potential problems with processing input/output files, it is recommended you limit the input/output file name to 30 characters or less.

It is recommended you put the input file in the specified directory. See [3.1 Import File Name and Location on page 14](#).

5.2.2 Time and Time Zones

Date and time elements referenced in Import Forecast refer to time in the CT's time zone.

Viewing the forecast in the *Adjust Forecast* view of the RCP for an entity containing the CT when the entity has a different time zone than the CT shows the data in the entity's time zone.

5.2.3 File Contents – General Specifications

◆ File Elements

The root element for Import Forecast is **forecastImport**.

5.2.4 Import Forecast DTD

The Import Forecast DTD contains the elements available to use in the Import Forecast XML file to specify the information you want to import.

Note: Do not create a DTD file for this import; only use this as a template for building XML input files.

```
<?xml version="1.0" encoding="UTF-8"?>
  <!ELEMENT forecastImport (ctForecast*)>
  <!ELEMENT ctForecast (ctID, forecastData+)>
  <!ELEMENT forecastData (date, (periodData+ | dailyData))>
  <!ELEMENT periodData (time, contactsReceived?, contactsActive?, absoluteAHT?)>
  <!ELEMENT dailyData (contactsReceived?, contactsActive?, (averageAHT |
absoluteAHT)?)>
  <!ELEMENT ctID (#PCDATA)>
  <!ELEMENT date (day, month, year)>
  <!ELEMENT contactsReceived (#PCDATA)>
  <!ELEMENT contactsActive (#PCDATA)>
  <!ELEMENT averageAHT (#PCDATA)>
  <!ELEMENT absoluteAHT (#PCDATA)>
  <!ELEMENT time (#PCDATA)>
  <!ELEMENT day (#PCDATA)>
  <!ELEMENT month (#PCDATA)>
  <!ELEMENT year (#PCDATA)>
  <!ATTLIST periodData
    recID NMTOKEN #IMPLIED>
  <!ATTLIST dailyData
    recID NMTOKEN #IMPLIED>
```

5.2.5 Import Forecast File Contents – Record Specifications

This table includes all elements available, from the Import Forecast DTD, to include in the Import Forecast XML file. See [5.2.4 Import Forecast DTD on page 46](#). The table consists of the element names, indented based on their level in the DTD, a description of the elements, and the valid values for the elements.

Field	Description	Valid Values
ctID	The NICE WFM CT numeric identifier for the forecast data to import.	0-999999999
forecastData	The date and type of data to import for the ctID .	n/a

Field	Description	Valid Values
date	The date for which the data is imported. Note: This must be later than the <i>date the import file is processed</i> . Past date forecast records cannot be modified.	n/a
day	The day-of-month part of a date element.	1-31
month	The month part of a date element.	1-12
year	The year part of a date element.	4-digit year
periodData	Indicates the data is for a single period. periodData and dailyData cannot be in the same date element for a ctID .	n/a
recID (optional attribute of periodData)	You may include a unique user-defined record identifier to aid in locating records.	n/a
time	The time period on the date for which data is imported, HH:MM, using 24-hour clock. Note: The time is the beginning of the associated time period.	00:00 - 23:45
contactsReceived	The periodic forecast contacts received for this date and time for the specified ctID .	0.00 - 65000.00
contactsActive	The periodic forecast contacts active for this date and time for the specified ctID .	0.00 - 65000.00
absoluteAHT	The periodic AHT value for the date and time for the specified ctID .	0.00 - 99999.00
dailyData	Indicates the data is for a daily total. periodData and dailyData cannot be in the same date element for a ctID .	n/a
recID (optional attribute of dailyData)	You may include a unique user-defined record identifier to aid in locating records.	n/a
contactsReceived	The daily forecast contacts received for this date for the specified ctID .	0.00 - 999999.00
contactsActive	The periodic forecast contacts active for this date for the specified ctID .	0.00 - 999999.00
averageAHT	The daily AHT value to use for the date for the specified ctID . This element is only valid for dailyData elements. Each period's AHT value for the date is adjusted so the resulting daily AHT value matches this imported value. Note: You cannot use averageAHT and absoluteAHT in the same dailyData element for a ctID .	0.00 - 99999.00
absoluteAHT	The AHT value used for each period of the day.	0.00 - 99999.00

Field	Description	Valid Values
	Note: You cannot use averageAHT and absoluteAHT in the same dailyData element for a ctID .	

5.3 Import Forecast XML Input File Example

This is an example of an Import Forecast XML file.

```
<?xml version="1.0" encoding="UTF-8" ?>
<forecastImport>
  <ctForecast>
    <ctID>200</ctID>
    <forecastData>
      <date><day>4</day><month>7</month><year>2021</year></date>
      <periodData recID="2001"><time>00:00</time>
        <contactsReceived>300</contactsReceived>
        <absoluteAHT>301</absoluteAHT>
      </periodData>
      <periodData recID="2002"><time>00:30</time>
        <contactsReceived>300</contactsReceived>
        <absoluteAHT>301</absoluteAHT>
      </periodData>
      <periodData recID="2003"><time>01:00</time>
        <contactsReceived>300</contactsReceived>
        <absoluteAHT>301</absoluteAHT>
      </periodData>
      <periodData recID="2004"><time>01:30</time>
        <contactsReceived>300</contactsReceived>
        <absoluteAHT>301</absoluteAHT>
      </periodData>
    </forecastData>
  </ctForecast>
  <ctForecast>
    <ctID>202</ctID>
    <forecastData>
      <date><day>4</day><month>7</month><year>2021</year></date>
      <dailyData recID="2021">
        <contactsReceived>2000</contactsReceived>
        <averageAHT>350</averageAHT>
      </dailyData>
    </forecastData>
    <forecastData>
      <date><day>5</day><month>7</month><year>2021</year></date>
      <dailyData recID="2022">
        <contactsReceived>2500</contactsReceived>
        <averageAHT>320</averageAHT>
      </dailyData>
    </forecastData>
    <forecastData>
      <date><day>6</day><month>7</month><year>2021</year></date>
      <dailyData recID="2023">
        <contactsReceived>2250</contactsReceived>
        <averageAHT>325</averageAHT>
      </dailyData>
    </forecastData>
  </ctForecast>

```

```

</ctForecast>
<ctForecast>
  <ctID>203</ctID>
  <forecastData>
    <date><day>4</day><month>7</month><year>2021</year></date>

```

```

    <dailyData recID="2031">
      <contactsReceived>2000</contactsReceived>
      <averageAHT>301</averageAHT>
    </dailyData>
  </forecastData>
</ctForecast>
</forecastImport>

```

5.4 Import Forecast Command Line

For distributed systems, including hosted/cloud environments, copy the import file onto the EFT site before executing the import forecast command. A process automatically moves the file from the EFT site to the server. It can take SmartSync 10 minutes or more to process files depending on the size of the files and the timing of tasks to move files.

You can execute the program manually or automatically on a pre-determined schedule.

Note: The Import Forecast input file is not deleted after a successful import, unless you run the `importForecast` command using the **-d** command line parameter. If the file cannot be deleted, this message appears in the output file: `Could not delete input file.`

5.4.1 Import Forecast Command Line Parameters

Import Forecast is invoked using the `importForecast` command and these parameters:

Parameter	Description
-c<custID> (required)	The ID of the customer partition for which to import the forecast. Valid values are 1 – 32757.
-f<path/file.name> (recommended)	Use this parameter to specify the path and name of the file to import. The recommended path and file name is <code>/totalview/ftp/imports/customer<custID>/forecasts/forecast.1.xml</code> If you do not provide a file parameter on the command line, the default is <code>/totalview/ftp/imports/forecast/forecast.<n>.xml</code> where <code><n></code> is the customer number.
-d<delete input file> (Y N) (optional)	Y - Delete the input file after processing it, even if there are errors when processing the file. N - Do not delete the file after processing it. If you do not provide this parameter, the file is not deleted after it is processed. Note: You are responsible for deleting the input files. Allowing input files to accumulate on the server may cause insufficient disk space issues.
-o<output file> (optional)	The path and name of an output file. Summary data is written to this file, including the number of elements processed, the number processed successfully, and the number that were not imported because of errors.

	If there are any errors, the Import Forecast process automatically creates an .ERR file. See 5.6 Import Forecast Output File on page 52 .
--	--

5.4.2 Import Forecast Command Line Example:

Refer to the *Administrator Guide - WFM* for details about running commands from the *Custom Activities* view.

Linux:

```
importForecast -c<custID> -  
f/totalview/ftp/imports/customer<custID>/forecasts/forecast.1.xml >  
/totalview/ftp/imports/customer<custID>/forecasts/run.log
```

Windows:

```
importForecast -c<custID> -  
f/totalview/ftp/imports/customer<custID>/forecasts/forecast.1.xml >  
/totalview/ftp/imports/customer<custID>/forecasts/run.log 2>&1
```

5.5 Import Forecast Processing

5.5.1 General Processing Information

Processing ends and an error message prints indicating the reason for the failure in these circumstances:

- you omitted the customer command line parameter, **-c**, or the specified customer ID is invalid.

The customer ID is considered invalid if either of these conditions is true:

- you did not specify a valid integer
- there is no customer defined in NICE WFM with the specified ID
- the file specified with the **-f** command line parameter does not exist

If a **ctForecast** element contains a child that is not allowed or you specified an invalid value in any child element whether the child is mandatory or not, the offending part of the **ctForecast** element is not imported, but the valid **forecastData** elements is imported. A message is included in the output file indicating the error. For example, if a **ctForecast** element contains seven **forecastData** elements a week, and one of the **forecastData** elements has an error but the others are correct, the other six **forecastData** elements are imported and the one with the error is logged to the error file without being imported.

For each **ctForecast** element, the program expects an active forecast to exist for the specified CT and date. You must generate an active forecast for the CT prior to the import. If an active forecast does not exist for the CT/date, a message is printed in the output file, including the entire **forecastData** element, and the record is not imported.

The **date** element must be later than the *date the import file is processed*. Past date forecast records cannot be modified.

For each updated period, including periods updated as a result of importing a daily value to distribute to periods within that date:

- Forecast requirement values are updated after the import values are written to the database. Forecast objectives stored in the **Forecast** record are used to calculate the forecast requirement values.
- For CTs which belong to a BU, allocation percentages for all CTs in the BU are calculated when any CT is imported.
- Requirements are not stored with multi-skill efficiency applied, so no processing for multi-skill efficiency is necessary.

Multi-skill efficiency values are not updated automatically by importing a forecast; therefore, you must run a simulation to update the multi-skill efficiency values.

For periods where the CT is not workload-based, **Contacts Handled** is set to 0.

For periods where the CT is workload-based, **Contacts Handled** is set to the same value as **contactsReceived**. Therefore, after the import file is processed, you must recalculate the **Contacts Handled** values by re-generating a plan (forecast) for the workload-based CT.

The *revised* data fields are updated, as opposed to the *original* data fields.

5.5.2 dailyData Elements

If you import a daily value rather than a value for each 15- or 30-minute period, the existing forecast data is used to distribute the daily values. CT hours are not honored; data is imported to periods which already have forecast data. If a period has 0 for the existing forecast value, that period still has a 0 after the daily values are distributed.

If you specified a **contactsReceived** value, it is distributed to the periods in the day based on the existing distribution of values within the day. The daily data is skipped if, after distribution, any period of the day receives more than 65,000 contacts received. For example, if the contacts received for a day is 200,000 and the active forecast of the day has only 2 non-zero periods, the daily data is not imported because at least one of the 2 periods receives more than 65,000 contacts.

If you specify an **averageAHT** value, each period's AHT is adjusted so the resulting daily AHT value matches this imported value.

If you specify an **absoluteAHT** value, each period's AHT is set to that value.

If you do not specify a value for **contactsReceived**, **averageAHT**, or **absoluteAHT**, the existing values for the date remain unchanged. For example, you can make use of this import to only import contacts received by not importing any AHT value and the existing AHT values remain unchanged in the forecast records.

5.5.3 perioddata elements

Using this element, you may import data for specific 15- or 30-minute time periods on specific dates. The **time** element is interpreted based on the corresponding CT's time zone. A **periodData** element does not allow an **averageAHT** element.

If you do not specify a value for **contactsReceived** or **absoluteAHT**, the existing value for the time period remains unchanged. For example, you can make use of this import to only import contacts received by not importing any AHT value and the existing AHT values remain unchanged in the forecast records.

5.5.4 CTs in a BU

Do not import data for **contactsReceived** for CTs assigned to a BU using **dailyData** elements. If a daily value is imported for one or more CTs in a BU, the data is distributed to the CT using each CT's intraday distribution. The CT-specific intraday distribution is a result of the allocation of contacts to each of the CTs. It does not necessarily reflect the intraday distribution of the BU. Therefore, the BU intraday distribution may be inappropriately modified after the import. Refer to this table for an example of what can occur after the import:

	BU		CT 1		CT 2				BU		
	Contacts	Intraday Distribution	Contacts	Intraday Distribution	Contacts	Intraday Distribution	CT 1	CT 2	Contacts	Intraday Distribution	Difference in
	<i>before</i>	<i>Percent</i>	<i>before</i>	<i>Percent</i>	<i>before</i>	<i>Percent</i>	<i>import of</i>	<i>import of</i>	<i>after</i>	<i>Percent</i>	BU Intraday
	import	import	import	import	import	import	100	500	import	import	Distribution
							<i>contacts</i>	<i>contacts</i>			<i>after</i>
											<i>import</i>
06:00	30	9%	0	0%	30	14%	0	70	70	12%	3%
06:30	30	9%	0	0%	30	14%	0	70	70	12%	3%
07:00	35	10%	0	0%	35	16%	0	81	81	14%	3%
07:30	40	12%	0	0%	40	19%	0	93	93	16%	4%
08:00	45	13%	35	27%	10	5%	27	23	50	8%	-5%
08:30	45	13%	30	23%	15	7%	23	35	58	10%	-3%
09:00	50	14%	20	15%	30	14%	15	70	85	14%	0%
09:30	70	20%	45	35%	25	12%	35	58	93	15%	-5%
<i>total</i>	<i>345</i>		<i>130</i>		<i>215</i>		<i>100</i>	<i>500</i>	<i>600</i>		

5.6 Import Forecast Output File

◆ -o parameter

If you use the **-o** command line parameter, an output file is created in the same directory as the import file. This output file provides summary information regarding the import processing.

Example:

```

<?xml version="1.0"?>
<forecastImport>

<!-- customer: 1 -->
<!-- file name: /totalview/ftp/imports/customer<custID>/forecasts/forecast.1.xml -->
<!-- output file name:
/totalview/ftp/imports/customer<custID>/forecasts/out.forecast.1.xml -->

<!-- summary data... -->
<!-- 120 forecastData elements were processed -->
<!-- 111 forecastData elements were successfully imported -->
<!-- 5 forecastData elements were not imported due to errors -->
<!-- 4 forecastData elements were not imported due to locked records -->
.

```

◆ Error File

If any errors occur during processing, an error file is created in the same directory as the input file, with the same name as the input file with a **.ERR** extension. Only records with errors are included in this file. Comments from the input file are not included in the error file.

The format of the error file is the same as the format of the input file so you can easily re-import the elements that caused errors after correcting the elements. All error messages and informative text, such as the customer ID, file name, summary data, and so on, are enclosed in XML comment tags. The first line of the output file is an XML declaration: `<?xml version="1.0"?>`. The contents of the error file are contained in a **ctForecast** root element.

Example .ERR file:

```

<?xml version="1.0"?>
<forecastImport>

<!-- customer: 1 -->
<!-- file name: /totalview/ftp/imports/forecast/forecast.1.xml -->

<!-- CT ID not a valid value -->
  <ctForecast>
    <ctID>123ABC</ctID>
    <forecastData>
      <date><day>4</day><month>7</month><year>2022</year></date>
      <dailyData><contactsReceived>2000</contactsReceived></dailyData>
    </forecastData>
  </ctForecast>

</forecastImport>

```

5.7 MU Requirements Conversion Utility

Outsourcers using NICE WFM are often provided interval-based requirement values from clients at the MU level and you must create schedules to cover the requirements. Often the requirement values are updated frequently, sometimes multiple times within a day. Therefore, the outsourcer needs the ability to process the provided requirement values in an automatic and frequent basis.

You cannot import requirement values using a SmartSync import because the system processes are built from the basic elements of a contact value, an AHT value, and a forecast objective. In a

Multi-skill Plan environment, more than just the requirement values are needed. AHT and contact volumes are also needed for simulation purposes.

As an alternative to importing the requirement values directly, a utility is available which accepts a file with requirement values. The utility converts the requirements values into the necessary contact values using a user-supplied AHT and statistics period length. The output of this utility includes:

- the contact volume for each interval for each CT
- user-supplied AHT for each CT
- a formatted file that can be imported using the SmartSync Import Forecast

5.7.1 MU Requirements Conversion Utility Input file

You must format the input file according to the SmartSync Data Export MU Forecast file. If the source of the MU requirements input file is another NICE WFM system, the file is easily produced using SmartSync Data Export. The fields must be pipe-delimited and are mandatory unless otherwise indicated. Fields must be in this order:

1. **date**, formatted as YYYYMMDD
2. **period**, formatted in 24-hour time
3. **time zone**, not used in the output file, only included in the input file to aid troubleshooting; this field may be null
4. Customer ID
5. Contact Type ID
6. **Contact Type name**, not used in the output file, only included in the input file to aid troubleshooting; this field may be null
7. **MU ID**, not used in the output file, only included in the input file to aid troubleshooting; this field may be null
8. **MU name**, not used in the output file, only included in the input file to aid troubleshooting; this field may be null
9. Requirement

◆ MU Requirements Conversion Utility Input File Example

```
20220104|08:00|CST6CDT|1|100|Sales|50|North Sales|25.00
20220104|08:30|CST6CDT|1|100|Sales|50|North Sales|25.00
20220104|09:00|CST6CDT|1|100|Sales|50|North Sales|30.00
20220104|09:30|CST6CDT|1|100|Sales|50|North Sales|30.00
20220104|10:00|CST6CDT|1|100|Sales|50|North Sales|30.00
```

◆ Input File Validations

The input file is validated. If any of the validations fail:

- the output file is not created
- an error file is created with the same name as the input file with **.ERR** appended to the end

- the original input file is not deleted, regardless of the command line option

The error file contains messages indicating the errors. Each invalid record is printed, with the error text appended to the end of the line, separated by an additional | symbol. **Note:** Valid records are not processed since no output file is created.

Validations are not processed to ensure fields such as **Customer ID** or **Contact Type ID** are within the valid range of IDs for that entity within NICE WFM, nor are validations performed to see if the values are defined on the destination server. When the file is imported with the SmartSync Import Forecast program, it validates the entities exist and creates error files as necessary.

5.7.2 MU Requirements Conversion Utility Command Line Parameters

You can execute the MU Requirement Conversion Utility program manually or automatically on a pre-determined schedule. The MU Requirements Conversion Utility may invoke the Import Forecast process. The utility uses `convReqToContacts` command and these parameters:

Parameter	Description
-c<custID> (required)	The customer ID for the SmartSync Import Forecast file.
-i<path/file.name> (required)	The path and name of the file to import.
-o<path/file.name> (required)	The path and name of the output file.
-a<AHT value> (required)	AHT value, in seconds, up to 2 decimal places.
-l<stat period length> (required)	Statistics period length in minutes, 15 or 30 .
-f<run Forecast Import> (optional)	Y - The SmartSync Import Forecast program automatically runs after the MU Requirements Conversion output file is created. N - The SmartSync Import Forecast program is invoked manually or through another cron-type mechanism. If you do not include this parameter, Import Forecast runs automatically after the MU Requirements Conversion output file is created.
-r<beginning record ID> (optional)	Beginning record ID to use in the output file.
-d<delete input file> (optional)	Indicates whether to delete the input file if successfully translated, Y or N. If you do not include this parameter, the input file is deleted after MU Requirements Conversion utility finishes successfully.
-E (optional)	Indicates an optional encoding value to include in the XML output file.

Example:

```
convReqToContacts -c1 -iinput.txt -ooutput.xml -a200.25 -l30 -fN -r1 -
dN -EUTF-8
```

5.7.3 MU Requirements Conversion Utility Output File

The output file is created from values in the input file, values from the command line, and values calculated by the conversion utility. The forecast contacts received for the output file are calculated by multiplying the requirements value by the statistics period length and dividing by the AHT. The resulting value is rounded to 2 decimal places, based on the .5 rule.

$$\text{Forecast Contacts} = [(\text{Requirements} * \text{StatisticsPeriodLength}) * 60] / \text{AHT}$$

The input records are not assumed to be in a particular order; therefore, it is not considered an error if the records are not sorted in a particular way. The output records are sorted by **ctID**, **date**, **time**.

It is possible for multiple CT IDs to be contained in the input file and these are reported appropriately in the output file. **Note:** It is not valid for there to be multiple records for a CT corresponding to different MU IDs in the input file.

◆ Output File Format

The output file is formatted to conform to Import Forecast specifications.

These elements are included:

```
ctForecast
ctID
forecastData
    date
        day
        month
        year
    periodData
        time
        contactsReceived (calculated)
        absoluteAHT (from the command line)
```

Note: The **dailyData** and **averageAHT** elements are not used.

If the optional record ID is provided on the command line, the first **periodData** element contains this **recID** parameter and it is incremented by 1 for each subsequent **periodData** element in this output file. If this optional record ID is not provided, the **recID** parameter is omitted from the **periodData** elements.

If the optional encoding value is provided, the XML declaration includes the encoding value.

◆ MU Requirements Conversion Output File Example

```
<?xml version="1.0"?>
<forecastImport>
    <ctForecast>
        <ctID>200</ctID>
        <forecastData>
            <date><day>14</day><month>2</month><year>2021</year></date>
```

```

<periodData recID="2018"><time>00:00</time>
  <contactsReceived>300</contactsReceived>
  <absoluteAHT>301</absoluteAHT>
</periodData>
<periodData recID="2019"><time>00:30</time>
  <contactsReceived>300</contactsReceived>
  <absoluteAHT>301</absoluteAHT>
</periodData>
<periodData recID="2020"><time>01:00</time>
  <contactsReceived>300</contactsReceived>

```

```

  <absoluteAHT>301</absoluteAHT>
</periodData>
<periodData recID="2021"><time>01:30</time>
  <contactsReceived>300</contactsReceived>
  <absoluteAHT>301</absoluteAHT>
</periodData>
</forecastData>
</ctForecast>
</forecastImport>

```

5.7.4 Source server setup

If the source server is a NICE WFM server, you can use the SmartSync Data Export MU Forecast module to create the input file.

Define the MU Forecast export to only include these fields in the output:

- **date** (YYYYMMDD format)
- **period** (24-hour format)
- **TZ**
- **custID**
- **ctID**
- **ctName**
- **muID**
- **muName**
- **revPlanReq**

Sort the fields by **custID**, **ctID**, **date**, **period**. The times exported should be relative to the time zone of the destination CT.

5.7.5 Destination server setup

The Import Forecast feature is required for this conversion program, since the importer is used to add the contacts to NICE WFM.

◆ Replicate the Requirements Value

If the goal is to *replicate* the requirements value on a destination server to match the values provided from a source server, you must define the CT on the destination server with one of these:

- A simulation (BerlangC) requirement calculation. The CT forecast objectives should be blank for **SL** (service level) and **ASA** and 100 for **MaxOcc** (maximum occupancy).
- A workload requirement calculation. The CT forecast objectives should be defined as 100% in 30 minutes, or 100% in 15 minutes, depending on the statistics period length).

◆ Translate the Requirements Value

If the goal is to *translate* the requirements value on a destination server to an *estimated* contact volume for a specified AHT with a specified **SL/ASA/MaxOcc** goal, you must define the CT on the destination server with simulation (BerlangC) as the requirement calculation method and the appropriate **SL/ASA/MaxOcc** goals. The requirement values on the destination CT do not necessarily match the requirement values on the source MU.

Note: No actual contact information, such as contacts offered, contacts handled, contacts abandoned, service level, ASA, and so on, is reported for the CT on the destination server. Therefore, the Intraday view on the destination server may have limited capability. Use the Forecast view or the MU views to view data on the destination server.

The time zone of the destination CT should match the times exported from the source server. No translation of times is performed to shift the times in the data files to match the time zone of the CT.

5.7.6 Limitations

If you define the CTs on the destination server as workload-based CTs, the CTs cannot belong to a BU. In addition, multi-skill efficiency is not applicable for workload-based CTs. These two limitations actually make sense for this type of integration since the intent is to store requirement values derived on an external system, and to generate schedules and manage staffing against those requirement values.

You are responsible for defining CTs on the destination server using the same CT IDs in the source server (input file). If CT IDs are already used on the destination server for other types of contacts, you need to modify the CT IDs in the input file to match the CT IDs on the destination server.

The CTs on the destination server must have the same time zone as the source CTs.

The source and destination entities must use the same statistics period length. Both must use either 15- or 30-minutes intervals.

6. Import History

Import History allows you to import CT historical data items from a standardized file. You can use the Import History feature:

- to reduce the data entry time required to populate queue history during the initial database build.
- to reduce the data entry time required to populate queue history as you define new queues.
- as an automatic data posting method to populate data from multimedia servers, email servers, fax servers, and so on.

Important Note: On distributed systems, including hosted/cloud environments, the former limitation against importHistory via Custom activities is removed in WFM 8.1+ because of scalable/HA swxiface. Provided the required shared filesystem for historical files is mapped to every WFM App server, Custom Activities can run importHistory.

Important Note: If you use this import, consider converting to the improved Import History XML Connector. It supports the import of agent queue statistics, agent system statistics, and agent activity details. For more information, contact NICE WFM Support.

This section covers these main topics:

- Information on where the imported data is stored
- The initial setup required before processing the import history file
- Import History file specifications for importing CT historical data into the WFM database
- Input file sample for importing CT historical data
- Command line parameters to run the Import History program
- Import History processing

Note: Implementation details to automate the import of CT historical data are not included.

6.1 Information for Import History

Data is imported into the **QUEUESTATS** table; you can see imported data on views such as *Queue History* and *Intraday*, as well as various reports.

Data is not imported into the **MUSTATS** table using the Import History program. This means these reports and views do not have data from queues or agents, if the data was populated using the Import History program:

- MU Results report
- Performance Analysis view and report
- MU Analysis
- MU Measurements

Data is not imported into **AGTQUEUESTATS**, **AGTSYSSTATS**, **AGTSKILLSTATS**, or **ACTIVITYLOGDTL** tables using the Import History program. The Import History program does not process agent signin/signout or real time data, so these views and reports do not show data for the contacts an agent handled on ACDs/servers that use the Import History program:

- Agent Detail view and report
- Queue Utilization report
- Adherence report
- Conformance report
- Real Time Adherence
- Agent State Summary

6.2 Initial Setup

Prior to processing an import file, define:

- a CT for each type of contact needed for forecasting and scheduling
- at least one queue for each CT
- Each ACD queue tag must match an **ID** in the import file

The CT is one of these types:

- BErlangC-based
- Workload-based

6.3 Steps after Upgrading from R7.x

If you use SmartSync Import History to import history for a CT that does **not** have at least one queue assigned, complete the following steps after the upgrade:

- Define a queue for each imported CT. The ACD queue tag must match an ID in the import file.
- Contact NICE WFM Support to restart your SmartSync Import History.

Note: If you use SmartSync Import History to import history for a CT that **does** have at least one queue assigned, contact NICE WFM Support to restart your SmartSync Import History after the upgrade.

6.4 Import History File Specifications

6.4.1 Import History File Name Specifications

Use one of these naming conventions for the file:

1. File name = `IMPORT.HIST` [NOT recommended for ongoing data feeds]
2. File name = `mmdyy.hmm` [recommended for ongoing data feeds]

To avoid potential problems with processing input/output files, it is recommended you limit the input/output file name to 30 characters or less.

6.4.2 Import History File Contents – General Specifications

The file can contain records for one period, a range of periods, one day, or any number of days.

Each record within the file provides summary data for a 15- or 30-minute increment. Each record has a date and timestamp.

You cannot combine data from multiple ACDs/servers in one file. All records in the import file must originate from the same ACD/server.

The **time** field in each record is the time according to the ACD/server's time zone, defined on the *ACD Parameters* view.

Shared time elements, such as, **ready** (available) time, **idle** time, and **login** time, must either be:

- Allocated, using your own method, to each **ID** in the file or
- included in only one **ID** in the file. In which case the **Estimated Staff** and **Actual Occupancy** values are not accurate.

The file contains a header record, one or more data records, and a trailer record.

- The header record is BEGIN CONTACT HISTORY IMPORT.
- The trailer record is END CONTACT HISTORY IMPORT.

Sort the records in the file by **ID**, then by **Date**, then by **Time**. If not, the processing time may be adversely affected.

An import file may not exceed 5 MB in size. If the input file is greater than 5MB, you must segment the file into smaller files.

6.4.3 Import History File Contents – Record Specifications

Each record within a file consists of fields delimited by double quotes and separated by commas. Additional formats may be acceptable; contact the NICE Professional Services group if a different delimiter is used.

Each record terminates with an ASCII new-line character, decimal 10.

Denote blank optional fields and filler fields by two double quotes with no characters between them.

Values do not have the thousands place separated by commas or any other delimiter.

The data fields for each record are listed below. You must include each field in each record in the order listed.

Note: You determine the formulas and source of these data items for each ACD or server. Therefore, it is incumbent on you to ensure the data calculations are meaningful and applicable to workforce management purposes.

Field	Description
ID (required)	1 to 40 characters; case is sensitive A queue of specific contacts routed to agents. The file record ID must be identical to an ACD queue tag in the queue definition defined in NICE WFM. The contents of this field can be numeric, alpha, or alphanumeric. Note: An ACD queue tag is associated with a CT in NICE WFM. This allows you to provide individual queue records in the file, and subsequently assign one or more queues to CTs in NICE WFM.
Date (required)	The date of the historical record you are reporting to NICE WFM. The valid formats are: ▪ YY/MM/DD ▪ YYYY/MM/DD ▪ MM/DD/YYYY
Time (required)	The time of the historical record for NICE WFM. The valid format is: ▪ HH:MM in 24-hour time ▪ the time must be in the ACD time zone This timestamp is used to determine the period in which to post the data. The time must be on the statistics period beginning boundary, for example, 12:00, 12:15, 12:30 or 12:45, which represent the period beginning time, and must match the statistics period length, 15 minutes or 30 minutes, as defined in NICE WFM.
Number of contacts received (required)	0-65000; integer The total number of contacts received in the period designated by the Date and Time above. If the data is for a BerlangC-based CT <u>and</u> if this field does not equal the sum of Number of Contacts Handled and Number of Contacts Abandoned, an error message prints in the log file and the value is imported as specified in the import file.
Number of contacts handled (required)	0-65000 The total number of contacts handled and completed in the period designated by the Date and Time above. For workload-based CTs, delayed response CTs such as emails, faxes, and so on, the Number of Contacts Handled may be greater than or less than the Number of Contacts Received in an interval.
ASA x 100 (optional)	0-999900, in seconds The average speed of answer for the contacts handled in the interval. Note: Since ASA is stored in NICE WFM to 2 decimal places, the value reported in the file must be multiplied by 100. This eliminates the need to parse the decimal indicator which can be either a period (.) or a comma (,).
AHT x 100 (optional)	0-9999900, in seconds The average handle time for contacts handled in the interval.

Field	Description
	This field is Filler. The system calculates the AHT as documented in the Contact Types section of the <i>Administrator Guide - WFM</i>.] Note: Since AHT is stored in NICE WFM to 2 decimal places, the value reported in the file must be multiplied by 100. This eliminates the need to parse the decimal indicator which can be either a period (.) or a comma (,).
Outbound contacts (optional)	0-65000 The total number of contacts generated by agents in response to the inbound contacts in the period designated by the Date and Time above. If you can associate each outbound contact with an inbound queue ID, the outbound contact information can be used for calculating loaded workforce requirements. Otherwise, this field is only for information purposes when reporting results in NICE WFM.
Number of contacts abandoned (optional)	0-65000 The total number of contacts abandoned in the period designated by the Date and Time above. For workload-based CTs, delayed response CTs such as emails, faxes, and so on, the Number of Contacts Abandoned may be greater than or less than the Number of Contacts Received in an interval. Note: This field may not be practical for workload-based CTs.
Number of contacts abandoned on or before service time (optional)	0-65000 The total number of contacts abandoned on or before the service level threshold. The value in this field should include contacts abandoned in the period designated by the Date and Time above. Define the threshold for each queue on the ACD, on the server, or in the report generator. Note: This field may not be practical for workload-based CTs.
Service level x 100 (required)	0-10000 The <i>percentage</i> of contacts handled and completed on or before the service level threshold. The value in this field should be based on contacts handled and completed in the period designated by the Date and Time above. Define the threshold for each queue on the ACD, on the server, or in the report generator. Therefore, if this number does not exist in your ACD or server databases, you must derive the value. Note: Since service level is stored in NICE WFM to 2 decimal places, the value reported in the file must be multiplied by 100. This eliminates the need to parse the decimal indicator which can be either a period (.) or a comma (,). Note: You must calculate service level and provide it as a percentage. Using Number of contacts handled on or before service time in an interval divided by Number of Contacts Received from the same interval, example formula below, is not usually accurate for delayed response media.

Field	Description
	$\frac{\text{ContactsHandledBeforeServiceObjectiveDuringPeriodX}}{\text{ContactsReceivedDuringPeriodX}}$ This is because it is possible these two values in a given interval are not related. For example, you may: - receive 1000 e-mails at 7:30am and not handle any at 7:30am - handle 500 e-mails at 8:00am and not receive any at 8:00am - receive 200 e-mails at 8:30am and handle 400, none of which were received at 8:30am, because the handled emails were the ones received at 7:30) Therefore, determine the service level percentage using one of these formulas: $\frac{\text{ContactsHandledBeforeServiceObjectiveDuringPeriodX}}{\text{ContactsHandledDuringPeriodX}}$ $\frac{\text{ContactsHandledBeforeServiceObjectiveDuringPeriodX}}{\text{ContactsHandledDuringPeriodX} + \text{ContactsHandledAfterServiceObjectiveDuringPeriodX}}$ $\frac{\text{ContactsHandledBeforeServiceObjectiveDuringPeriodX}}{\text{ContactsHandledDuringPeriodX} + \text{ContactsAbandonedDuringPeriodX}}$
Filler	This field is reserved for future use.
In time (required)	0-9999999, in seconds The total number of seconds agents spent on contacts that were handled and completed. The value in this field should include contacts handled and completed in the period designated by the Date and Time above. This is often referred to as talk time for inbound calls or work time for emails.
Out time (optional)	0-9999999, in seconds The total number of seconds agents spent on outbound contacts, see Outbound contacts above, in the period designated by the Date and Time above. If you can associate each outbound contact with an inbound queue ID, the outbound contact information can be used for calculating loaded workforce requirements. Otherwise, this field is only for information purposes when reporting results in NICE WFM.
Filler	This field is reserved for future use.
Ready time (optional)	0-9999999, in seconds The total number of seconds agents were: - logged into the system AND - ready and waiting/available for contacts AND - not actively talking or working on contacts in the period designated by the Date and Time above.
Work time	0-9999999, in seconds

Field	Description
(optional)	The total number of seconds agents spent on contacts that were handled and completed. The value in this field should include contacts handled and completed in the period designated by the Date and Time above. This is often referred to as after call work time for inbound calls or research time for emails.
Idle time (optional)	0-9999999, in seconds The total number of seconds agents were: ▪ logged into the system AND ▪ <u>not</u> ready, <u>not</u> waiting, <u>not</u> available for contacts AND ▪ not actively talking or working on contacts in the period designated by the Date and Time above.
Login time (optional)	0-9999999, in seconds The total number of seconds agents were logged into the ACD or server. The value in this field should include time logged in to the period designated by the Date and Time above.
Backlog not yet expired (optional)	0-65000 The total number of contacts received in previous intervals that have not been worked, completed, handled, or abandoned by the end of the designated Date and Time above. These contacts can still be worked in future periods. The service threshold has <u>not</u> passed for these contacts. Note: This field is normally used for workload-based CTs only.
Backlog expired (optional)	0-65000 The total number of contacts received in previous intervals that have not been worked, completed, handled, or abandoned by the end of the designated Date and Time above. These contacts can still be worked in future periods. The service threshold has passed for these contacts. Note: This field is normally used for workload-based CTs only.
Active Contacts (optional)	0-65000 The number of contacts reported by the ACD that were worked during the interval but received in some prior interval. Note: This field is not used for BUs or for Outbound OCM CTs.

◆ Distribution of Common Data Elements

These data items are typically not reported by the ACD/server for each CT or queue:

- Outbound Contacts
- Out Time
- Ready Time
- Idle Time
- Login Time

Some of these data items are used by NICE WFM to calculate loaded AHT, estimated staff, and actual occupancy for each CT. If the ACD/server is not able to report these data items by queue, you can:

- develop an algorithm in the file generation program to allocate shared data items to each queue or
- ignore estimated staff values, actual occupancy values, or disable loaded AHT within NICE WFM.

6.5 Import History Input File Example

```
BEGIN CONTACT HISTORY IMPORT
"1","10/26/2020","8:00","49","43","","","5","","","3400","","74","","","174","","",""
"1","10/26/2020","8:30","49","43","","","5","","","3400","","74","","","174","",""
"1","10/26/2020","9:00","72","62","","","10","","","2100","","95","","","559","",""
"1","10/26/2020","9:30","72","62","","","10","","","2100","","95","","","559","",""
"1","10/26/2020","10:00","97","92","","","5","","","6800","","57","","","724","",""
"1","10/26/2020","10:30","97","92","","","5","","","6800","","57","","","724","",""
"1","10/26/2020","11:00","84","81","","","3","","","7300","","67","","","734","",""
"1","10/26/2020","11:30","84","81","","","3","","","7300","","67","","","734","",""
"1","10/26/2020","12:00","69","66","","","2","","","9100","","73","","","556","",""
"9","5/10/2021","18:00","1","1","","","0","","","10000","","1294","","","1","",""
"9","5/11/2021","10:00","5","5","","","0","","","8000","","330","","","0","",""
"9","5/11/2021","10:30","1","1","","","0","","","10000","","13","","","89","",""
"9","5/11/2021","11:00","3","3","","","0","","","10000","","143","","","160","",""
"9","5/11/2021","12:00","2","2","","","0","","","10000","","783","","","407","",""
"9","5/11/2021","13:00","3","3","","","0","","","6700","","315","","","0","",""
"9","5/11/2021","13:30","1","1","","","0","","","10000","","256","","","160","",""
"9","5/11/2021","15:00","1","1","","","0","","","10000","","345","","","0","",""
"9","5/11/2021","15:30","1","1","","","0","","","10000","","587","","","149","",""
"9","5/12/2021","13:30","2","0","","","2","","","0","","","0","","","0","",""
END CONTACT HISTORY IMPORT
```

6.6 Import History Command Line

For distributed systems, including hosted/cloud environments, copy the import file onto the EFT site before executing the import history command. A process automatically moves the file from the EFT site to the server. It can take SmartSync 10 minutes or more to process files depending on the size of the files and the timing of tasks to move files.

You can execute the program manually or automatically on a pre-determined schedule.

Note: The Import History input file is deleted after a successful import, unless you run the `importhistory` command using the `-K` command line parameter. If the file cannot be deleted, this message appears in the output file: `Could not delete input file.`

6.6.1 Import History Command Line Parameters

Import History is invoked using the `importhistory` command and these parameters:

Parameter	Description
-c<custID> (required)	The ID of the customer partition for which the agents are imported. This value must be an integer and must not be zero padded.
-a<ACD ID> (required)	The ACD ID in NICE WFM. This is used by the program to identify the ACD/server from which the data in the file was collected. Create one file

	per ACD or server. Each file is processed separately by running the Import History program for each file.
-AC	Use this parameter in the command to enable and process the ActiveContact field in the Import History file. This is an optional field. If this parameter is not given, the existing Import History file without the ActiveContact field is processed.
-d<path/file.name> (recommended)	Use this parameter to specify the path and name of the file to import. The recommended path and file name is /totalview/ftp/imports/customer<custID>/history/IMPORT.HIST If you do not provide a file parameter on the command line, the default is /ftp/switches/<acdID>
-E (optional)	The .ERR output file is not written.
-f<form> (optional)	The form filename in /usr/prms/run/conf (default is form.<ACD ID>. For example, form.21).
-o (optional)	Overwrite existing data. If you omit this parameter, the program does NOT overwrite periods for which data exists in the database.
-K (optional)	Keep the imported data file after processing completes. If you omit this parameter, the import file is deleted after processing completes.

6.6.2 Import History Command Line Example

Refer to the *Administrator Guide - WFM* for details about running commands from the *Custom Activities* view.

Linux:

```
importhistory -c<custID> -a<ACD ID> -  
d/totalview/ftp/imports/customer<custID>/history/IMPORT.HIST -o >  
/totalview/ftp/imports/customer<custID>/history/IMPORT.OUT
```

Windows:

```
importhistory -c<custID> -a<ACD ID> -  
d/totalview/ftp/imports/customer<custID>/history/IMPORT.HIST -o >  
/totalview/ftp/imports/customer<custID>/history/IMPORT.OUT 2>&1
```

6.7 Import History Output file

An output file is created with error messages and statistics regarding the number of records imported and processed. The output file has the same name as the input file with the **.ERR** extension. Possible error messages are:

Validation	Error message
User attempts to import data with a record which has too many or too few fields.	Line: <line no>: Incorrect number of fields: <incorrect_number_of_fields>.
The ACD Queue Tag exists in the record, but the ACD Queue Tag does not exist in NICE WFM.	Line: <line no>: Queue Tag <Queue Tag> does not exist in NICE WFM.

Validation	Error message
User attempts to import data for a date and time that is the current or a future period.	Line: <line no.>: Data for the current and future periods may not be imported; current date and time is <YYYY/MM/DD> <HH:MM>. <i>Where the date and time are included in the message.</i>
The user attempts to import data for a date and time earlier than the beginning of the existing history file for that CT.	Line: <line no.>: Data for <date> <time> not imported since current history begins on <history_start_date>.
The ACD Queue Tag is blank.	Line: <line no.>: Blank Queue Tag: <invalid_Queue_Tag>.
The date is blank.	Line: <line no.>: The date is a required field.
The date contains invalid characters.	Line: <line no.>: Invalid characters in date: <invalid_date>.
The time is blank.	Line: <line no.>: The time is a required field.
The time contains invalid characters.	Line: <line no.>: Invalid characters in time: <invalid_time>.
The time is not on a statistics period boundary.	Line: <line no.>: Time must be on a <15 or 30>-minute boundary: <invalid_time> is not valid. <i>Where either 15 or 30 is printed, based upon the associated statistics period.</i>
The contacts received value is not greater than or equal to 0 and less than or equal to 65000.	Line: <line no.>: Contacts received must be greater than or equal to 0 and less than or equal to 65000: <invalid_value> is not valid.
The contacts handled value is not greater than or equal to 0 and less than or equal to 65000.	Line: <line no.>: Contacts handled must be greater than or equal to 0 and less than or equal to 65000: <invalid_value> is not valid.
The contacts handled before service time value is not greater than or equal to 0 and less than or equal to 65000.	Line: <line no.>: Contacts handled before service time must be greater than or equal to 0 and less than or equal to 65000: <invalid_value> is not valid.
The outbound contacts value is not greater than or equal to 0 and less than or equal to 65000.	Line: <line no.>: Outbound contacts must be greater than or equal to 0 and less than or equal to 65000: <invalid_value> is not valid.
The contacts abandoned value is not greater than or equal to 0 and less than or equal to 65000.	Line: <line no.>: Contacts abandoned must be greater than or equal to 0 and less than or equal to 65000: <invalid_value> is not valid.
The contacts abandoned before service time value is not greater than or equal to 0 and less than or equal to 65000.	Line: <line no.>: Contacts abandoned before service time must be greater than or equal to 0 and less than or equal to 65000: <invalid_value> is not valid.
The service level is not between 0.00 and 10000.	Line: <line no.>: The Service Level must be greater than or equal to 0.00 and less than or equal to 10000: <invalid_value> is not valid.
The <X> time is not greater than or equal to 0 and less than or equal to 2147483647, where <X> can be In, Out, Ready, Work, Idle, Login.	Line: <line no.>: <X> time must be greater than or equal to 0 and less than or equal to 2147483647: <invalid_value> is not valid.

Validation	Error message
The backlog not yet expired value is not greater than or equal to 0 and less than or equal to 65000.	Line: <line no.>: Backlog not yet expired must be greater than or equal to 0 and less than or equal to 65000: <invalid_value> is not valid.
The backlog already expired value is not greater than or equal to 0 and less than or equal to 65000.	Line: <line no.>: Backlog already expired must be greater than or equal to 0 and less than or equal to 65000: <invalid_value> is not valid.
The CT being imported is not WORKLOAD and Number of Contacts Received does not equal the sum of Number of Contacts Handled and Number of Contacts Abandoned .	Line: <line no.>: Number of contacts received does not equal the sum of Number of contacts handled plus Number of contacts abandoned. Value imported as specified anyway.
Number of active contacts is less than 0 or greater than 65000.	Line: <line no.>: Active Contacts must be greater than or equal to 0 and less than or equal to 65000.

The last lines of the error file indicate various statistics about the number of records in the file. This example illustrates possible statistics if you chose the option not to overwrite existing data:

- 96 total records in file
- 4 records with errors
- 14 records not imported because data already existed
- 78 records successfully imported

If you chose the option to allow data to be overwritten, these statistics display:

- 96 total records in file
- 4 records with errors
- 14 records overwritten
- 92 records successfully imported

7. Import MU Opens

Import MU Opens allows the import of interval-based MU Scheduled Open values into NICE WFM for one or more MUs.

- The Import MU Opens feature is useful for collecting and displaying scheduled open data from outsource vendors, without needing to see the actual agent schedules which drive the interval values for scheduled open.
- The imported values are stored as external opens, separate from scheduled open values derived from NICE WFM agent schedules.
- You can view the external open values on some MU screens. The sum of the external opens and scheduled opens derived from NICE WFM agent schedules is shown on CT screens.
- You can import multiple MUs in the same data file.

This section covers these main topics:

- Import MU Opens file specifications for importing interval-based MU Scheduled Open values into the WFM database.
- Input file sample for importing MU Scheduled Open values.
- Command line parameters to run the Import MU Opens program.
- Import MU Opens processing, including information on the error file.
- Import MU Opens DTD.

Note: Implementation details to automate the import of MU Scheduled Open data are not included.

7.1 Import MU Opens File Specifications

7.1.1 File Name and Location Specifications

To avoid potential problems with processing input/output files, it is recommended you limit the input/output file name to 30 characters or less.

You should put the input file in the specified directory. See [3.1 Import File Name and Location on page 14](#).

7.1.2 Time and Time Zones

Date and time elements referenced in Import MU Opens refer to time in the MU's time zone.

7.1.3 File Contents – General Specifications

◆ File Elements

The root element for Import MU Opens is **opens**.

7.1.4 Import MU Opens DTD

Note: Do not create a DTD file for this import; only use this as a template for building XML input files.

```
<!ELEMENT opens (muOpens*) >
<!ELEMENT muOpens (muID, ctID?, date, openValue*)>
<!ELEMENT muID (#PCDATA)>
<!ELEMENT ctID (#PCDATA)>
<!ELEMENT date (day, month, year)>
<!ELEMENT openValue (#PCDATA)>
<!ELEMENT day (#PCDATA)>
<!ELEMENT month (#PCDATA)>
<!ELEMENT year (#PCDATA)>

<!--
!ATTLIST openValue time CDATA #REQUIRED-->
```

7.1.5 Import MU Opens File Contents – Record Specifications

This table includes all elements available, from the Import MU Opens DTD, to include in the Import MU Opens XML file. See [7.1.4 Import MU Opens DTD on page 71](#). The table consists of the element names, indented based on their level in the DTD, a description of the elements, and the valid values for the elements.

Field	Description	Valid Values
opens	The root element. This element can contain zero or more muOpens children.	
muOpens	This element contains all the opens data for a single MU, CT, and date.	Must include one muID child, one ctID child (optional), one date child, and zero or more openValue children.
muID	The MU ID. The specified CT must belong to the same EG as the MU, or the CT must be blank or -1.	An MU must exist in NICE WFM with this ID for the customerID in the command line.
ctID	The CT ID. This field can also be blank or -1. In both cases, -1 is stored. No error message is logged in this case.	A CT must exist in NICE WFM with this ID for the customerID in the command line.
date	The date for which the open value is stored.	This must be a valid date, using day , month , and year children. See 7.4 Import MU Opens Processing on page 74 for date validations.
day	The day of the month the open value is stored.	1-31
month	The month the open value is stored.	1-12
year	The year the open value is stored.	Any year
openValue	The number of agents open for the MU and the CT at the specified date and time period.	0.0000-9999.0000 (integers or float).

Field	Description	Valid Values
time	An attribute of the openValue element. This is the beginning time of the 15-minute period for which the open value is stored. It is assumed to be in the MU's time zone.	24-hour format, on a 15-minute increment, must be 00:00-23:45.

7.2 Import MU Opens Input File Example

This is an example of the Import MU Opens XML file.

Example:

```
<?xml version="1.0" encoding="UTF-8-1" ?>
<opens>
  <muOpens>
    <muID>28</muID>
    <ctID>27</ctID>
    <date>
      <day>17</day><month>5</month><year>2021</year>
    </date>
    <openValue time="08:00">12.00</openValue>
    <openValue time="08:15">12.00</openValue>
    <openValue time="08:30">12.00</openValue>
    <openValue time="08:45">12.00</openValue>
    <openValue time="09:00">12.00</openValue>
    <openValue time="09:15">12.00</openValue>
    <openValue time="09:30">12.00</openValue>
    <openValue time="09:45">12.00</openValue>
  </muOpens>
  <muOpens>
    <muID>28</muID>
    <ctID>28</ctID>
    <date>
      <day>17</day><month>5</month><year>2021</year>
    </date>
    <openValue time="08:00">54.00</openValue>
    <openValue time="08:15">52.00</openValue>
    <openValue time="08:30">55.00</openValue>
    <openValue time="08:45">57.00</openValue>
    <openValue time="09:00">59.00</openValue>
    <openValue time="09:15">59.00</openValue>
    <openValue time="09:30">62.00</openValue>
    <openValue time="09:45">60.00</openValue>
  </muOpens>
  <muOpens>
    <muID>28</muID>
    <ctID>-1</ctID>
    <date>
      <day>17</day><month>5</month><year>2021</year>
    </date>
    <openValue time="08:00">2.00</openValue>
    <openValue time="08:15">1.00</openValue>
    <openValue time="08:30">0.00</openValue>
    <openValue time="08:45">1.00</openValue>
    <openValue time="09:00">3.00</openValue>
  </muOpens>
</opens>
```

```

        <openValue time="09:15">2.00</openValue>
        <openValue time="09:30">4.00</openValue>
        <openValue time="09:45">5.00</openValue>
    </muOpens>
    <muOpens>
        <muID>28</muID>
        <ctID>27</ctID>
        <date>
            <day>18</day><month>5</month><year>2021</year>
        </date>
        <openValue time="08:00">5.00</openValue>
        <openValue time="08:15">5.00</openValue>
        <openValue time="08:30">4.00</openValue>
        <openValue time="08:45">5.00</openValue>
        <openValue time="09:00">6.00</openValue>
        <openValue time="09:15">6.00</openValue>
        <openValue time="09:30">9.00</openValue>
        <openValue time="09:45">9.00</openValue>
    </muOpens>
    ...
</opens>

```

7.3 Import MU Opens Command Line

For distributed systems, including hosted/cloud environments, copy the import file onto the EFT site before executing the import MU opens command. A process automatically moves the file from the EFT site to the server. It can take SmartSync 10 minutes or more to process files depending on the size of the files and the timing of tasks to move files.

You can execute the program manually or automatically on a pre-determined schedule.

7.3.1 Import MU Opens Command Line Parameters

Import MU Opens is invoked using the `importOpens` command and these parameters:

Parameter	Description
-c<custID> (required)	The ID of the customer partition for which to import the MU Opens. Valid values are 1-32757.
-f<path/file.name> (optional)	Use this parameter to specify the path and name of the file to import. The recommended path and file name is /totalview/ftp/imports/customer<custID>/schedules/opens.1.xml If you do not provide a file parameter on the command line, the default path and file name is /ftp/imports/opens/opens.xml.

7.3.2 Import MU Opens Command Line Example

Refer to the *Administrator Guide - WFM* for details about running commands from the Custom Activities view.

Linux:

```

importOpens -c<custID> -
f/totalview/ftp/imports/customer<custID>/schedules/opens.1.xml >
/totalview/ftp/imports/customer<custID>/schedules/run.log

```

Windows:

```
importOpens -c<custID> -  
f/totalview/ftp/imports/customer<custID>/schedules/opens.1.xml >  
/totalview/ftp/imports/customer<custID>/schedules/run.log 2>&1
```

7.4 Import MU Opens Processing

Processing ends and an error message prints indicating the reason for the failure if the Import MU Opens feature is not enabled for the specified customer

7.4.1 General Processing Information

Each **muOpens** element contains all the opens data for a single MU, CT, and date. You can include periods for which the open value is zero, but they are not required. Any period in the date for which you do not include data is set to zero. You can include data for schedules that cross midnight with the date on which the schedule starts or include it with the next date's data. (The NICE WFM MU Opens export includes cross-midnight data with the date of the schedule.)

Note: You are responsible for deleting the input files. Allowing input files to accumulate on the server can cause insufficient disk space issues.

You can sort records in the input file in any order.

The importer only changes the MU's schedule date range if the MU belongs to an EG on the new date AND if the validations listed below pass.

Start date validations:

- The start date cannot be more than 581 days earlier than the current date.
- Plan/forecast data must exist for the MU on the start date.

End date validations:

- The end date must be on or after the SchedRange record's start date.
- Plan/forecast data must exist for the MU at least two days after the end date.

7.4.2 Error File

If there are any errors during processing, an error file is created in the same directory as the input file, with the same name as the input file with a **.ERR** extension. Only records which had errors are included in this file.

8. Import Schedule Changes

Import Schedule Changes automatically updates one or more agents' schedules by applying an activity code for a specified date and time.

- You provide activity code and other pertinent data in a specifically formatted file.
- The schedule change may come from an IVR system, a web browser, a third party platform, and so on.
- You can add to existing schedules or fully overwrite an existing schedule.
- You can create future activities using this import.

You can manually invoke the program and it searches a specified directory for files to process, including any files of records that could not be processed due to locked tables on a previous attempt. You can use functionality in NICE WFM and UNIX to automatically schedule the invocation of this program at specified times.

Invalid records are logged to an error file. Records that cannot be modified because of locked tables are stored and the modifications are repeatedly attempted until a specified timeout is reached.

This section covers these main topics:

- Mapping activity codes.
- Import Schedule Changes file specifications for updating agent schedules in the WFM database.
- Input file samples for importing agent schedule changes.
- Importing agent schedule changes using the Knowlagent/Intradiem interface.
- Command line parameters to run the Import Schedule Changes program.
- Import Schedule Changes processing, including information on command errors, activity code priorities, formatting, creating future activities, deleting schedule events and future activities, the lock file, and the output files.

Note: Implementation details to automate the import of schedule changes data are not included.

8.1 Information for the Import Schedule Changes Program

8.1.1 Pipe-delimited File Restrictions

These command line parameters are not supported **when the input file is pipe-delimited**:

- **-o:** This parameter must not be used. This parameter indicates schedules are completely overwritten with the imported activities.
- **-N:** This parameter must not be used. This parameter indicates the acknowledgement (**ACK**) file contains no more than the specified maximum records.

8.2 Activity Code Mapping

For a schedule event in the import file to be mapped to an activity code in the WFM database, you must define an activity code attribute and assign a value for the attribute to activity codes used to update the schedules. The attribute values need to be the same as the alphanumeric text in the import file, see [exception on page 78](#).

The default attribute is *SchedExImport*. Define this attribute and assign values for this attribute to the activity codes used to represent the imported schedule events. You can designate a different attribute in the command line parameters, see [8.6 Import Schedule Changes Command Line on page 87](#).

This is an example of how to define the attribute used for mapping schedule events in the import file to the activity codes on the WFM server.

Step 1: Define Activity Code Attribute

MENU: ADMINISTRATOR > ACTIVITY CODE ATTRIBUTES

- Create an activity code attribute for each of the customer partitions on each WFM server.
- The name of the activity code attribute must be *SchedExImport* or it can be user defined.
- For each schedule event included in the import file, enter an **Attribute Value**. For example, assume the import file has Training, Meeting, Tardy, Vacation, and Coaching as text descriptors of the scheduled events.

The screenshot shows the 'Activity Code Attributes' window for 'Multi-Skill Customer'. The main table lists various attributes, with 'SchedExImport' highlighted. Below it, the 'List values' window shows the values assigned to 'SchedExImport'.

Description	Data type	Decimal places	Minimum value	Maximum value	Currency symbol	List
MC Group	Alphanumeric					☒
Meeting Sch...erwritable	Logical					☒
ML Esthetically Pleasing	Alphanumeric					☒
Paid Time Off - DB	Logical					☐
Revenue Generating RS	Logical					☐
SchedExImport	Alphanumeric					☒
Schedule Codes BM	Alphanumeric					☒
Variance-No OPA	Alphanumeric					☐
Variance-OPA	Alphanumeric					☐

Values
Coaching
Meeting
Tardy
Training
Vacation

Step 2: Assign Attribute Values to Activity Codes

MENU: ADMINISTRATOR > ATTRIBUTE ASSIGNMENT

For each *SchedExImport* attribute value defined, select one activity code to represent the event.

Example:

Icon	Description	SchedExImport	Schedule Codes BM	Variance-No OPA	Variance-OPA
	GB-Military Leave		M		
	GB-On Loan		O		
	GB-Personal Day		P		
	GB-Personal Time		W		AUX
	GB-Travel		T		
	Goofing Off -RM		G		
	Happy Hour - BM		H		Logout
	Holiday Closed		H		Logout
	Holiday Open		H		
	Jury Duty				
	Leave Early				
	LP Admin		A		
	LP Side by Side Coaching		C		
	LS Documentation		D		
	LS Peer Coaching		C		
	Lunch		L		Lunch
	MC Service Email		E		
	Meeting	Meeting	M		Logout
	ML-Fire Drill				Logout
	Open				Login
	Overtime		O		Login
	Personal Day		P		Logout
	Project Mode				
	Sick Paid		S		Logout
	Sick Unpaid		S		Logout
	SV Coaching				
	SV Sabbatical				
	Tardy	Tardy			Login
	Training	Training	T		Logout
	Vacation	Vacation	V		Logout
	Voting Day				

The selected activity code for an attribute value appears on the schedule after the import file is processed. This mapping concept allows you to send event descriptors which are not specifically defined as activity codes in the WFM database.

8.3 Import Schedule Changes File Specifications

8.3.1 File Name and Location Specifications

The file name uses the format `ts.id` where `ts` is a timestamp indicating the file creation date/time (Unix time format – in seconds (10 digits)) and `id` is an alphanumeric identifier indicating the system that created the file. For example, `1627153200.IFC` indicates a file was created at 2:00pm on 7/24/2021 by a system referred to as IFC. The file name extension can be any 3 alphanumeric characters with the exception of: `LCK`, `ERR`, `ACK`, `KIT`.

8.3.2 Date, Time and Time Zones

Time elements in Import Schedule Changes refer to the agent's MU time zone.

These examples refer to *not* using the `-o` parameter. Only schedule changes are imported, rather than the complete schedule, see [8.6 Import Schedule Changes Command Line on page 87](#).

The **StopTime** can be earlier than the **StartTime**, which indicates it crosses midnight. The activity only applies to the schedule that starts on the indicated date.

Example:

If an agent works from 20:00 (8:00pm) to 06:00 on Monday through Friday, and a record has a time range of 01:00-03:00 for a Wednesday-dated record, the 01:00-03:00 time range actually applies to the calendar date of Thursday but is applied to the schedule with the Wednesday date.

Example:

If an agent does not have a schedule for a Monday, and is scheduled on Tuesday from 3:00 until 12:00, and a record has a Monday date with a time range of 23:00 to 05:00, the record is not processed. Even though it appears the time range crosses midnight from Monday into Tuesday and falls onto the Tuesday schedule, since the record date is a Monday, only the Monday schedule is modified and in this example there was no Monday schedule.

8.3.3 File Contents – General Specifications

The input file can be a pipe-delimited text file or an XML formatted file. Contact NICE Support to ensure your version of NICE WFM supports the XML format.

If you create the input file on a Windows PC and transmit it to the WFM server using FTP, the type should be set to ASCII to ensure proper translation of characters such as end-of-line, carriage return and line feed. Transmitting the file as binary can create a situation where ^M characters exist at the end of each record preventing the file from being processed.

8.3.4 File Contents – Pipe-Delimited File Record Specifications

If you use a pipe-delimited file format, each record contains these fields. A record contains 3, 4, 5, 6, 7, or 8 fields. Field order must be as specified. If you omit an optional field, you must include a place holder (|) if you include one of the other optional fields.

agentID|exception|schedule date|startTime|endTime|delete|otCreateSched|exception date

Parameter	Description
agentID (required)	The agent's personal ID or the NICE WFM agent ID in the WFM database. - If you use the -iA command line parameter: The value can have a maximum of 9 digits and cannot contain non-numeric characters. - If you use the -iS command line parameter: Values specified in id elements must be unique personal ID numbers. The value can be 256 alphanumeric characters.
exception (required)	The user-specified schedule event. The value in this field must be defined as a value in the <i>SchedExImport</i> activity code attribute. You can specify a different activity code attribute as a command line parameter. Using an activity code attribute allows you to easily control the mapping of an event to an activity code. Note: The Unscheduled (grey icon) activity code should not have a value assigned for the selected activity code attribute. Note: This is the same as the code field in XML input files, see the next section.

Parameter	Description
schedule date (required)	The date on which the schedule containing the activity starts. The schedule date can be different than the exception date, if the schedule crosses midnight. The format is YYYYMMDD.
startTime (optional)	The start time of the activity code. The format is HH:MM, 24-hour clock. A blank field indicates Start of Schedule, where the system identifies the agent's schedule start time.
endTime (optional)	The stop time of the activity code. The format is HH:MM, 24-hour clock. A blank field indicates End of Schedule, where the system identifies the agent's schedule end time. The endtime is the clock stop time, and therefore excludes the stop minute. For example, if a Lunch activity exists at noon for one hour, the endtime is 1:00pm, not 12:59pm.
Delete (optional)	If the word delete is present in this field, the specified activity is deleted from the agent's schedule and replaced by the base schedule code and deleted from the Future Activity table if a corresponding record exists. Possible values are blank or delete. Not case sensitive. Note: If any text other than delete is found in this field, an error message is logged; the activity is not deleted.
otCreateSched (optional)	If this field is populated, the type of future activity created is <i>OT Partial Day</i> and the <i>Create Schedule</i> flag is set for the future activity. Possible values include blank or create schedule. Not case sensitive. Note: Only use this field if the -f or -F command line parameter is present. If this field is populated and you omit the -f or -F parameter, a warning message is logged. The activity is imported to the schedule; however, the future activity record is not created. Note: The type of future activity created is <i>OT Partial Day</i> if this field is populated. However, a start and end time are both required for an <i>OT Partial Day</i> future activity. Therefore, if you omit either startTime or endTime, the future activity created is not <i>OT Partial Day</i>, regardless of the setting in this field. ▪ If you omit startTime, the record type is <i>Partial Day Sched Start</i> ▪ If you omit endTime, the record type is <i>Partial Day Sched End</i> ▪ If you omit both times, the record type is <i>All Day</i> Note: You cannot use this with vacation-related future activities.
exception date (optional)	The date on which the activity starts. The schedule date can be different than the exception date, if the schedule crosses midnight. If you do not include this field, the schedule date is used as the exception date. The format is YYYYMMDD.

8.3.5 Import Schedule Changes File Contents – XML File Specifications

◆ File Elements

The root element for Import Schedule Changes is **agentScheduleList**.

◆ Import Schedule Changes DTD

The DTD specifies the elements to include in the XML file.

Note: Do not create a DTD file for this import; only use it as a template for building XML input files.

```
<?xml version="1.0" encoding="UTF-8" ?>
<!ELEMENT agentScheduleList (agent+)>
<!ELEMENT agent (id, schedule)>
<!ELEMENT schedule (schedid?, day, month, year, (exception+ | offexception))>
<!ELEMENT exception (startdate, starttime?, endtime?, code,
    vacationGroup?, vacationYear?, note?,
    agentAllowedViewNote?,
    modifiedBy?, requestedBy?, timeRequested?,
    delete?, otcreatesched?)>

<!ELEMENT offexception EMPTY>
<!ELEMENT startdate (day, month, year)>
<!ELEMENT starttime (hour, min)>
<!ELEMENT endtime (hour, min)>
<!ELEMENT id (#PCDATA)>
<!ELEMENT schedid (#PCDATA)>
<!ELEMENT code (#PCDATA)>
<!ELEMENT vacationGroup (#PCDATA)>
<!ELEMENT vacationYear (#PCDATA)>
<!ELEMENT note (#PCDATA)>
<!ELEMENT agentAllowedViewNote (#PCDATA)>
<!ELEMENT modifiedBy (#PCDATA)>
<!ELEMENT requestedBy (#PCDATA)>
<!ELEMENT timeRequested (day, month, year, hour, min)>
<!ELEMENT delete EMPTY>
<!ELEMENT otcreatesched EMPTY>
<!ELEMENT day (#PCDATA)>
<!ELEMENT month (#PCDATA)>
<!ELEMENT year (#PCDATA)>
<!ELEMENT hour (#PCDATA)>
<!ELEMENT min (#PCDATA)>
```

◆ Import Schedule Changes XML Record Specifications

This table includes all elements available, from the Import Schedule Changes DTD, to include in the Import Schedule Changes XML file. See [Import Schedule Changes DTD on page 80](#). The table consists of the element names, indented based on their level in the DTD, a description of the elements, and the valid values for the elements.

Field	Description	Valid Values
agentScheduleList	The root element.	

Field	Description	Valid Values
agent	The element used to identify each agent and the agent's schedule change.	
id	The ID of the agent. The contents of this field is determined by the -i flag used on the program command line. See 8.6 Import Schedule Changes Command Line on page 87 .	If -iA is used: 1-999999999 If -iS used: The value may be 256 alphanumeric characters.
schedule	The schedule element; contains these five child elements and one or more exception elements.	
schedid	Use this field to assign a unique ID to the import record. It is included in the response file and lock file.	
day	The day of the month in which the schedule occurs.	1-31
month	The month of the year in which the schedule occurs.	1-12
year	The year on which the schedule occurs.	Any year; 4 digits
offexception	This element does not contain any data; it is a flag that indicates to delete the schedule. Note: This element requires the -o command line parameter. If an offexception exists in the file followed by an exception element for the same agent and date, the exception is stored. If an exception element exists in the file followed by an offexception for the same agent and date, no schedule is stored. Note: This field is not available for pipe-delimited input files, see previous section.	
exception	There are one or more exception elements to indicate the required change to the agent's schedule; each contains a startdate , starttime , endtime , and code child elements.	
startdate	The date on which the activity begins.	
day	The day of the month on which the activity begins.	1-31
month	The month of the year in which the activity begins.	1-12
year	The year in which the activity begins.	Any year; 4 digits
starttime	The time the activity begins. If you omit this field, Start of Schedule is used, where the	

Field	Description	Valid Values
	system identifies the agent's schedule start time.	
hour	The hour the activity begins, in 24-hour format.	0-23
min	The minute the activity begins.	0-59
endtime	The time the activity ends. The endtime is the clock stop time, and therefore excludes the stop minute. For example, if a Lunch activity exists at noon for one hour, the endtime time is 1:00pm, not 12:59pm. If you omit this field, End of Schedule is used, where the system identifies the agent's schedule end time.	
hour	The hour the activity ends, in 24-hour format	0-23
min	The minute the activity ends.	0-59
code	This data item contains the user specified schedule event. You must define the value in this field as a value in the activity code attribute used on the command line. Using an activity code attribute allows you to easily control the mapping of an event to an activity code. Note: This is equivalent to the exception field in pipe-delimited input files. See 8.3.4 File Contents – Pipe-Delimited File Record Specifications on page 78.	
vacationGroup	The ID of the time off group for which the future activity record is stored. Note: No validations are performed to determine whether or not the agent is assigned to the specified time off group. Note: This is only available with XML input files; it cannot be used with pipe-delimited input.	1-2147483647 A time off group must be defined in NICE WFM with this ID.
vacationYear	The name of the time off rule for which the future activity record is stored. Note: This is only available with XML input files; it cannot be used with pipe-delimited input.	Blank or any printable ASCII characters except & and \. A time off rule must exist with this name for the specified time off group. Case is sensitive.
note	The note stored with the future activity and the scheduled event.	Blank or any printable ASCII characters except &

Field	Description	Valid Values
	Note: This is only available with XML input files; it cannot be used with pipe-delimited input.	and \. Maximum length is 256.
agentAllowedViewNote	This element indicates if the note is viewable by the employee. If you use the note element and the agentAllowedViewNote element is either absent, or included with no value or any value other than <i>false</i> , the note is viewable by the employee.	True False
modifiedBy	The logon name of the supervisor that last modified the future activity record. Note: This is only available with XML input files; it cannot be used with pipe-delimited input.	Blank or any printable ASCII characters except & and \. Maximum length is 256. Case is sensitive.
requestedBy	The logon name of the supervisor that initially submitted the vacation request for the agent. A blank value in this field indicates the request was made by the agent. Note: This is only available with XML input files; it cannot be used with pipe-delimited input.	Blank or any printable ASCII characters except & and \. Maximum length is 256. Case is sensitive.
timeRequested	The date and time the vacation request was initially submitted. Note: This is only available with XML input files; it cannot be used with pipe-delimited input.	Blank or any date and time, using day , month , year , hour , and min children
day	The day of the month on which the activity was requested.	1-31
month	The month of the year in which the activity was requested.	1-12
year	The year in which the activity was requested.	Any year; 4 digits
hour	The hour in which the activity was requested.	0-23
min	The minute in which the activity was requested.	0-59
delete	This element does not contain any data; it is a flag that indicates to delete the activity and replace it with the base schedule code. If you run the import with the -f or -F command line parameter, the corresponding future activity record is also deleted, if one exists.	

Field	Description	Valid Values
	Note: If you include the -o command line parameter, this field is not used.	
otcreatesched	This element does not contain any data; it is a flag indicating to create an OT future activity. Note: If you include the -o command line parameter, this field is not used. Note: You cannot use this with vacation-related future activities. Note: Only use this element if you include the -f or -F parameter. If you include this element and omit the -f or -F parameter, a warning message is logged. The activity is still imported to the schedule; however, the future activity record is not created. Note: The type of future activity created is <i>OT Partial Day</i> if this element is present. However, a <i>start and end time are both required</i> for an <i>OT Partial Day</i> future activity. Therefore, if you omit either the startTime or the endTime, the future activity created is not <i>OT Partial Day</i>, regardless of the presence of this element. ▪ If you omit the startTime, the record type is <i>Partial Day Sched Start</i>. ▪ If you omit the endTime, the record type is <i>Partial Day Sched End</i>. ▪ If you omit both times, the record type is <i>All Day</i>.	

8.4 Import Schedule Changes Input File Examples

8.4.1 Pipe-Delimited Input File

```

432192388|Tardy|20210730||9:00
432192388|Tardy|20210730||10:00
123451341|Tardy|20210730||10:30
29357195|Tardy|20210730||8:00
349593874|Personal|20210730|08:00|12:00
349873471|Tardy|20210730||14:00
459613845|Tardy|20210730||8:30
39573234|Personal|20210730|13:00|
459613845|Tardy|20210730||8:30
321945143|Tardy|20210730||8:30
342534554|Open|20210730|02:30|04:00|||20210730
342534554|Break|20210730|04:00|04:15|||20210730

```

8.4.2 XML Input File

```
<?xml version="1.0" encoding="UTF-8" ?>

<agentScheduleList>
  <agent>
    <id>470185</id>
    <schedule>
      <day>12</day>
      <month>8</month>
      <year>2021</year>
    <exception>
      <startdate>
        <day>12</day>
        <month>8</month>
        <year>2021</year>
      </startdate>
      <starttime>
        <hour>22</hour>
        <min>0</min>
      </starttime>
      <endtime>
        <hour>2</hour>
        <min>0</min>
      </endtime>
      <code>Open</code>
    </exception>
    <exception>
      <startdate>
        <day>13</day>
        <month>8</month>
        <year>2021</year>
      </startdate>
      <starttime>
        <hour>2</hour>
        <min>0</min>
      </starttime>
      <endtime>
        <hour>3</hour>
        <min>0</min>
      </endtime>
      <code>Lunch</code>
    </exception>
    <exception>
      <startdate>
        <day>13</day>
        <month>8</month>
        <year>2021</year>
      </startdate>
      <starttime>
        <hour>3</hour>
        <min>0</min>
      </starttime>
      <endtime>
        <hour>7</hour>
        <min>0</min>
      </endtime>
    </exception>
  </agent>
</agentScheduleList>
```

```

        </endtime>
        <code>Open</code>
    </exception>
</schedule>
</agent>
<agent>
    <id>470185</id>
    <schedule>
        <day>13</day>
        <month>8</month>
        <year>2021</year>
        <offexception/>
    </schedule>
</agent>
</agentScheduleList>

```

8.5 Knowagent/Intradiem Interface

The Knowagent on-line learning system creates and sends files to NICE WFM to erase learning breaks from agent schedules canceled by the Knowagent system, permitting the adherence reporting in NICE WFM to be more accurate.

The specification of the Import Schedule Changes feature as described in this document applies to the Knowagent interface, with the additional details in this section outlining the specifics for this interface.

8.5.1 Knowagent Interface Setup

To interface to the Knowagent on-line learning system:

- you must create a directory on the WFM server, for example, `/ftp/import/kiface`
- you must create an activity code attribute, for example, **KIFACE**. See [8.2 Activity Code Mapping on page 76](#). A single value of **Open** is created for this attribute. This value is assigned to the **Open** activity code. No other activity codes are assigned to this or any other value in the **KIFACE** attribute column.

8.5.2 Knowagent File name Convention

For the file name, use the format `ts.KIT` where `ts` is a timestamp indicating the file creation date/time in Unixtime format; `KIT` is a shorthand form of Knowagent Interface to WFM. For example, a file name of `1627153200.KIT` indicates a file that was created at 2:00pm on 7/24/2021 by the Knowagent system.

8.5.3 Knowagent Interface Command Line

An example command line for invoking the program to process files from the Knowagent system is:

```
schedeximport -c<cust_id> -d"/ftp/import/kiface" -a"KIFACE" [-t<timeout>] -iA
```

Customer ID, timeout value, and agent ID type are determined for each customer and specified when this integration is configured by the NICE Professional Services team.

8.5.4 Knowagent Example Input File

In this example, the first two agents were in learning breaks that were canceled at 7:56. The next two agents were in learning breaks that were canceled at 8:07. The last two agents were in learning breaks that were canceled at 8:18. The file indicates that the agents' schedules are changed to the Open activity code for the portion of the learning break not delivered.

```
1234|Open|20210730|07:56|08:00
1236|Open|20210730|07:56|08:00
5462|Open|20210730|08:07|08:15
4978|Open|20210730|08:07|08:15
2144|Open|20210730|08:18|08:30
1255|Open|20210730|08:18|08:30
```

8.6 Import Schedule Changes Command Line

For distributed systems, including hosted/cloud environments, copy the import file onto the EFT site before executing the import schedule changes command. A process automatically moves the file from the EFT site to the server. It can take SmartSync 10 minutes or more to process files depending on the size of the files and the timing of tasks to move files. If you are creating future activities using this import, it can take 20 minutes or more to process the files.

You can execute the program manually or automatically on a pre-determined schedule.

Note: The Import Schedule Changes input file is deleted after a successful import, unless you run the `schedeximport` command using the **-k** command line parameter. If the file cannot be deleted, this message appears in the output file: `Could not delete input file.`

8.6.1 Import Schedule Changes Command Line Parameters

Import Schedule Changes is invoked using the `schedeximport` command and these parameters:

Parameter	Description
-c<custID> (required)	The ID of the customer partition for which the agents are imported. Valid values are 1-32757.
-d<directory> (optional)	Use this parameter to specify the path to the folder to import. If you do not provide a folder parameter on the command line, the default path is <code>\$WORK/SchedExImport</code> .
-a<attribute name> (optional)	The activity code attribute to use when processing the schedule changes. The default value is SchedExImport .
-i<id type> (optional)	The type of ID used to identify agents: S - personal ID A - NICE WFM agent ID The default is S . You must specify unique values in id elements.
-t<timeout> (optional)	The timeout value in minutes.

Parameter	Description
	0 to 9999 The default is 120.
-o (optional)	<i>This option is not available when you use pipe-delimited input files.</i> Each agent's existing schedule is deleted for the date before data is imported. Before a schedule is deleted, all activities associated with the schedule to import in its place are validated to ensure the import is a valid activity code. If any invalid activities exist for an agent, the schedule changes are not imported for that agent. Note: Activities are not necessarily contiguous, for example, in the case of a split shift, however, activities should not overlap. If two activities do overlap, the activity with the higher activity code priority takes precedence. If the overlapping activities have the same priority, the order of the activities in the file determines which is overwritten. That is, the first activity is overwritten by the second. Note: You cannot use the -o parameter with the -f or -F parameter. If you include the -o and -f or -F parameters on the command line, an error is logged and the import is not run.
-x (optional)	The Import Schedule Changes import file is in XML format. In addition, the program's output is in XML format. Note: If the input file is pipe-delimited, the output file is in XML format even though the -x parameter is omitted.
-f (optional)	For each activity code imported, except for Open [blue icon] and Unscheduled [grey icon] activity codes, a corresponding future activity record is also created in addition to modifying the schedule. Note: The -F command line parameter indicates to only create future activity records. In addition, when an activity code is deleted, as designated by the optional delete field or element in the input file – see 8.7.10 Deleting Schedule Events & Future on page 98, if a future activity exists that corresponds to the deleted activity code, the future activity is also deleted. Note: You cannot use the -f parameter with the -o or -F parameters. If you include the -f and -o or -F parameters on the command line, an error is logged and the import is not run. Note: You can only import vacation future activities if you use the -x parameter; meaning XML input format is required.
-F (optional)	For each imported activity code, except for Open [blue icon] and Unscheduled [grey icon] activity codes, a future activity record is created, <i>but no change is made to existing schedules</i>. Note: The -f command line parameter indicates to update schedules in addition to creating future activity records. In addition, when you include the optional delete element in the input file, see 8.7.10 Deleting Schedule Events & Future on page 98, if a future activity exists that corresponds to the deleted activity, it is deleted.

Parameter	Description
	Note: You cannot use the -F parameter with the -o or -f parameters. If you include the -F and -o or -f parameters on the command line, an error is logged and the import is not run. Note: You can only import vacation future activities if you use the -x parameter; meaning XML input format is required.
-L<logname> (optional)	The log name is stored in the schedule record for all schedules updated by the import
-N<max agts in ACK file> (optional)	XML input files only. The acknowledgement (ACK) file does not contain more than the specified maximum records. If the maximum is exceeded, the ACK file is broken into several files. The ACK file is used for a custom interface. <i>This option is not available if you use pipe-delimited input files.</i>
-k (optional)	The input files are not deleted after processing the import. If the parameter is omitted, the input files are deleted after processing.

8.6.2 Import Schedule Changes Command Line Examples

Refer to the *Administrator Guide - WFM* for details about running commands from the Custom Activities view.

◆ Import Schedule Changes – Add Activity

Linux:

```
schedeximport -c<custID> -
d/totalview/ftp/imports/customer<custID>/schedules/add_activity/ -iA -x >
/totalview/ftp/imports/customer<custID>/schedules/add_activity/run.log
```

Windows:

```
schedeximport -c<custID> -
d/totalview/ftp/imports/customer<custID>/schedules/add_activity/ -iA -x >
/totalview/ftp/imports/customer<custID>/schedules/add_activity/run.log 2>&1
```

◆ Import Schedule Changes – Delete Activity

Linux:

```
schedeximport -c<custID> -
d/totalview/ftp/imports/customer1/schedules/del_activity/ -iA -x >
/totalview/ftp/imports/customer1/schedules/del_activity/run.log
```

Windows:

```
schedeximport -c<custID> -
d/totalview/ftp/imports/customer1/schedules/del_activity/ -iA -x >
/totalview/ftp/imports/customer1/schedules/del_activity/run.log 2>&1
```

◆ Import Schedule Changes – Delete Schedule

Linux:

```
schedeximport -c<custID> -  
d/totalview/ftp/imports/customer<custID>/schedules/del_schedules/ -iA -o -x >  
/totalview/ftp/imports/customer<custID>/schedules/del_schedules/run.log
```

Windows:

```
schedeximport -c<custID> -  
d/totalview/ftp/imports/customer<custID>/schedules/del_schedules/ -iA -o -x >  
/totalview/ftp/imports/customer<custID>/schedules/del_schedules/run.log 2>&1
```

◆ Import Schedule Changes – Future Activity

Linux:

```
schedeximport -c<custID> -  
d/totalview/ftp/imports/customer1/schedules/fut_activity/ -iA -x >  
/totalview/ftp/imports/customer1/schedules/fut_activity/run.log
```

Windows:

```
schedeximport -c<custID> -  
d/totalview/ftp/imports/customer1/schedules/fut_activity/ -iA -x >  
/totalview/ftp/imports/customer1/schedules/fut_activity/run.log 2>&1
```

8.7 Import Schedule Changes File Processing

Processing ends and an error message prints indicating the reason for the failure if the Import Schedule Changes feature is not enabled for the specified customer

8.7.1 General Processing Details

◆ Shortened or Extended Schedules

If the -o parameter is not used:

Records imported by this process do not extend schedules. For example, if a schedule is from 8:00am to 5:00pm and a record is imported for overtime from 5:00pm to 7:00pm, the record is not applied and the appropriate message is logged to the error file.

Records imported by this process do not shorten schedules. For example, if a schedule is from 8:00am to 5:00pm and a record is imported to apply Unscheduled time from 1:00pm to 5:00pm, the record is not applied and the appropriate message is logged to the error file. **Note:** A record is imported and applied for an activity such as Vacation or Meeting; this limitation only applies to overwriting existing schedule time with the Unscheduled activity.

If the -o parameter is used:

The -o parameter allows you to delete the entire schedule and rebuild the schedule using the import record. In this manner, you can shorten or lengthen a schedule, but be aware the entire schedule is deleted prior to the import of the new schedule activities. If the import record contains only a portion of the original schedule, the result may not be what you intended.

Note: The **-o** parameter is not available if using pipe-delimited input files.

Note: Some of the processing edits on the following pages which relate to date and time validations are not used when you use the **-o** parameter.

◆ Date Outside MU Date Range

If the **-o** parameter is not used:

The import process does not create schedules on days where schedules do not exist for an agent's MU. If the date of a record is not within the valid schedule range for the MU associated with the agent in that record, the record is not processed and an error message is logged:

```
ERROR: The date is not in the valid range YYYYMMDD-YYYYMMDD for MU muid
```

where *YYYYMMDD-YYYYMMDD* is the valid date range for the MU associated with the agent for this record and *muid* is the MU ID associated with this agent.

If the **-o** parameter is used:

This processing only takes place if the command line includes the **-o** parameter.

Note: The **-o** parameter is not available if using pipe-delimited input files.

The **-o** parameter indicates existing schedules are deleted and overwritten with the imported schedule.

- If schedule changes are sent to the NICE WFM system for a date that falls farther in the future than the MU's schedule date range, the date range is extended, if possible.
- If no schedule date range record exists for the MU, one is created. The start date of the new record's range is the current date. The end date is the latest date for which data is imported for the MU.
- If a schedule date range record does exist for the MU, it is updated so the *enddate* is the latest date for which data is imported for the MU.

Whether modifying an existing schedule date range record or creating a new record, the *enddate* is validated before it is changed:

- The end date must be on or after the schedule date range record's start date.
- Forecast data must exist for the MU at least two days after the end date.

If the end date, the latest date for which data is imported for the MU, is invalid, the schedule date range record is not updated. In this case, the schedule activity records imported for dates on and after the existing schedule date range fail.

Note: Some of the processing edits on the following pages which relate to date and time validations are not used when the **-o** parameter is used.

◆ Attribute Column Does Not Exist

Command Line Specified Attribute Column

If you specify an attribute column on the command line that does not exist, no records are processed; all records are logged to the error file with the message:

```
ERROR: Attribute <name> is specified but does not exist.
```

where <name> is the attribute name specified.

Default Attribute Column

If you do not specify an attribute column, *SchedExImport* is used. If this attribute does not exist, no records are processed; all records are logged to the error file with the message:

```
ERROR: "SchedExImport" attribute does not exist, and no alternate is specified.
```

◆ No Values Defined For Attribute Column

If you either use the default attribute column or specify an attribute column to use, and no values are defined for that attribute column, no records are processed; all records are logged to the error file with the message:

```
ERROR: Attribute <name> exists but has no values defined.
```

where <name> is the attribute name specified.

◆ Incorrect Number of Fields in Data Record

If you specify too many or too few fields for a record, the record is not processed and an error message is logged:

```
ERROR: Too many fields.
```

or

```
ERROR: Too few fields.
```

◆ Id Type is S and Personal ID Not Found

If the ID type is **S** and no agent is found with the indicated personal ID, the record is not processed and an error message is logged:

```
ERROR: SSN <n> not found.
```

where <n> is the specified personal ID value.

◆ Id Type is S and Invalid Personal ID

If the ID type is **S** and the specified personal ID is invalid, the record is not processed and an error message is logged:

```
ERROR: SSN <n> is not valid.
```

where <n> is the specified personal ID value.

◆ Id Type is S and More Than One Agent With the Same Personal ID

If the ID type is **S** and more than one agent has the indicated personal ID, the record is not processed. An error message is logged:

```
ERROR: This SSN value found for Agent:MU <a>:<n> and <b>:<m>.
```

where *<a>* is the first agent ID and *<n>* is that agent's associated MU ID and ** is the second agent ID and *<m>* is that agent's associated MU ID. If additional agents also have the personal ID value, the message contains additional *id:MU* text to indicate all agents with the personal ID value.

◆ **Id Type is A and Agent Id Not Found**

If the ID type is **A** and no agent is found with the indicated NICE WFM agent ID, the record is not processed and an error message is logged:

```
ERROR: Agent Id <n> not found.
```

where *<n>* is the specified agent ID value.

◆ **Id Type is A and Invalid Agent Id**

If the ID type is **A** and the specified agent ID is less than or equal to 0 or it contains non-numeric characters, the record is not processed and an error message is logged:

```
ERROR: Agent Id <x> is not valid.
```

where *<x>* is the specified agent ID value.

◆ **Activity Not Found**

If the activity value included in the import file is not defined as a value for the indicated attribute column, the record is not processed and an error message is logged:

```
ERROR: Exception not found for attribute <AttributeName>.
```

where *Exception* is from the input record and *<AttributeName>* is the attribute column used for mapping to an activity code.

◆ **Activity Maps to More Than One Activity Code**

If the activity value is mapped to more than one activity code for the indicated attribute column, the record is not processed and an error message is logged:

```
ERROR: Exception maps to more than one Exception.
```

where *Exception* is from the input record.

8.7.2 Activity Code Priorities

Activity code priorities are honored. Higher-priority activities are not overwritten. Equal and lower priority activities are overwritten

Note: This processing differs from the current method in the Apply Future Activities processing, which reverts to an internal ID as the tie-breaker when attempting to apply an equal priority activity to another.

Partial records can be applied and an error message is logged. For example, if the record indicates an activity of Tardy from 08:00 to 10:00, but a higher-priority activity already exists from 09:00-10:00, only 08:00-09:00 is overwritten with the Tardy activity. The error message is:

```
WARNING: HH:MM-HH:MM not applied because higher priority exception
<exception_name> exists.
```

where *<exception_name>* is the activity code description of the activity already on the schedule with a higher priority.

Special processing is necessary for the Open activity and the Unscheduled activity. **Note:** The processing here does not refer to activities with the **Open** attribute checked; the Open activity described here is the predefined Open activity code.

◆ Applying a record whose Activity field maps to the Open Activity

The Open activity code is always the highest priority activity when applied to a schedule. That is, if the record being applied is the Open activity, it overwrites any other activity on the schedule except for the Unscheduled activity, which is the absence of an activity.

For example, if a schedule has a Meeting activity from 3:00pm to 5:00pm and a record with the Open activity is applied from 4:00pm to 5:00pm, the Open activity overwrites the Meeting activity, regardless of the priority of the Meeting activity.

◆ Applying a record to a schedule which contains the Open Activity

When an activity is applied to the Open activity, the Open activity is always overwritten. For example, a schedule has the Open activity from 1:00pm to 5:00pm and a record with a Personal Time activity is applied from 4:00pm to 5:00pm, the Open activity is overwritten regardless of the priority of the Personal Time activity.

◆ Applying a record whose Activity field maps to the Unscheduled Activity

This case is not currently permitted. If the activity maps to the Unscheduled activity, that record is not processed and an error message is logged:

```
ERROR: Exception maps to Unscheduled.
```

where *Exception* is from the input record.

◆ Applying a record to a schedule which contains the Unscheduled Activity

Normal (non-split) shifts

Processing in [8.7.7 Start/stop times of record fall completely outside schedule on page 96](#) and [8.7.8 Start/stop times of record fall partially outside schedule on page 96](#) alerts you that records were not applied to the unscheduled time preceding or following a normal (as opposed to split) shift.

Note: The database records do not actually store the unscheduled time preceding and following a normal shift (schedule).

Split shifts

The unscheduled time between shifts in a split shift is stored. Records are not applied to this unscheduled time, much in the same manner as described in [8.7.7 Start/stop times of record fall](#)

[completely outside schedule on page 96](#) and [8.7.8 Start/stop times of record fall partially outside schedule on page 96](#).

If the time values in the record fall completely during the unscheduled portion of a split shift, the record is not processed and an error message is logged:

```
WARNING: The specified times fall completely during the unscheduled
portion of the split shift for agent <agent_id>.
```

where <agent_id> is the agent ID.

If the time values in the record overlap unscheduled time, for example, if the split shift schedule has the first shift from 08:00 to 12:00 and the second shift from 18:00-22:00, and the record indicates a time range of 10:00 to 14:00, only the portion of the record that overlaps non-unscheduled time is processed. In this example, the activity applies to 10:00-12:00.

```
WARNING: HH:MM-HH:MM not applied to the unscheduled portion of the
split shift.
```

where *HH:MM-HH:MM* is the range where the activity is not applied. In this example, it is 12:00-14:00. As another example for this split shift schedule, if the record specified the time as 10:00 to 20:00, there is a warning message referring to 12:00-18:00 not being applied and the activity is applied from 10:00-12:00 and from 18:00 to 20:00.

8.7.3 Date formatting

For a pipe-delimited file, format the date as YYYYMMDD. If the date field is not formatted in this way, the record is not processed and an error message is logged:

```
ERROR: Invalid date: <date>.
```

where <date> is the date field from the record.

8.7.4 Start and Stop time formatting

For a pipe-delimited file, format the times as HH:MM, or blank. If a time is not formatted in this way, the record is not processed and an error message is logged:

```
ERROR: Invalid time: <time>.
```

where <time> is the date field from the record.

8.7.5 Start and Stop times

The **StartTime** is optional. If not provided, the activity is applied from the start of the schedule up to the provided **EndTime**.

The **EndTime** is optional. If not provided, the activity is applied from the indicated **StartTime** up to the end of the schedule.

Both **StartTime** and **StopTime** can be blank, in which case the activity is applied to the entire schedule for the provided **Date**; activity code priorities are still honored.

8.7.6 Agent does not have schedule for specified date

If you don't use the **-o** parameter and an agent does not have a schedule for the specified date, the record is not processed and an error message is logged:

```
WARNING: Agent <agent_id> does not have a schedule for the specified date.
```

where *<agent_id>* is the agent ID.

Note: The **-o** parameter is not available if using pipe-delimited input files.

8.7.7 Start/stop times of record fall completely outside schedule

If you don't use the **-o** parameter and an agent has a schedule but the time values in the record are completely outside of the schedule start/end time, the record is not processed and an error message is logged:

```
WARNING: The schedule for agent <agent_id> does not contain any  
portion of the specified times.
```

where *<agent_id>* is the agent ID.

Note: The **-o** parameter is not available if using pipe-delimited input files.

8.7.8 Start/stop times of record fall partially outside schedule

If you don't use the **-o** parameter and an agent has a schedule but the time values in the record overlap unscheduled time, for example, if the schedule is from 08:00 to 17:00 and the record indicates a time range of 07:00 to 10:00, only the portion of the record that overlaps scheduled time is processed (in this example, the activities apply to 08:00-10:00).

```
WARNING: HH:MM-HH:MM not applied because of schedule range.
```

where *HH:MM-HH:MM* is the range where the activity is not applied. In this example, it is 07:00-08:00.

Note: The **-o** parameter is not available if using pipe-delimited input files.

8.7.9 Creating Future Activities

These elements are specific to future activity records, with the noted caveat; each element is optional. In addition, these elements do not affect any changes made to schedules, with the noted caveat.

- **vacationGroup**
- **vacationYear**
- **note** (You can use the **note** element for schedule changes and any future activities.)
- **modifiedBy**
- **requestedBy**
- **timeRequested**
- **otcreatesched**

Therefore, if any of these elements, with the exception of the **note** element, are present but you do not use the **-f** or **-F** command-line parameters, a warning message is logged.

◆ Vacation Future Activities

If you include either a **vacationGroup** or a **vacationYear** element, you must include both elements. Otherwise, an error is logged and the future activity record is not created.

Since a vacation future activity cannot be an overtime activity, if you include the **vacationGroup** and **vacationYear** elements in addition to the **otcreatesched** field, an error is logged and the future activity record is not created.

If any future activity records are created or deleted whose **vacgrp** and **vacyear** fields are populated, the *Time Off Bidding* views are automatically updated.

◆ Using the -f parameter

If the import program is run with the **-f** command-line parameter, for each activity code in the import file that maps to a valid NICE WFM activity code other than Open or Unscheduled, a future activity is created.

If the import record is valid, the future activity is created, regardless of whether you added the activity to the agent's schedule. For example, even if an activity cannot be applied to an agent's existing schedule for the entire time range specified because of existing higher-priority activities or because the schedule times do not contain the entire activity time specified, the future activity record is still created for the exact time range specified in the import record.

The future activity record is created even if no schedule exists on the specified date for the agent. This is true whether the agent is scheduled for a day off on the specified date or whether the specified date is outside of the MU's existing schedule range.

◆ Using the -F parameter

If you use the **-F** command-line parameter, processing is the same as the **-f** parameter, except no changes are made to schedules for any of the imported activities.

◆ Viewing the imported Future Activities

Using the *Future Activities* view, you can view, modify, or delete the created future activity record. The future activity fields, as viewed on the *Future Activities* view, are populated as:

Field	Description
Icon	The icon of the activity code to which the imported activity is mapped, using the specified activity code attribute.
Description	The description of the activity code to which the imported activity is mapped, using the specified activity code attribute.
Start date	The activity date specified in the import record. It is subject to cross-midnight and other validations, if you use the pipe-delimited file format option.
End date	The activity date specified in the import record. It is subject to cross-midnight and other validations, if you use the pipe-delimited file format option.
Day of Week	The day of the week on which the activity date falls is enabled.

Type	These types are created, based on the import data items: ▪ <i>Partial day</i>: if you provide both the starttime and endtime and omit the otcreatesched field or element ▪ <i>Partial day sched start</i>: if you provide the endtime and not the starttime ▪ <i>Parital day sched end</i>: if you provide the starttime and not the endtime ▪ <i>All day</i>: if you don't provide both the starttime and endtime ▪ <i>OT partial day</i>: if you provide both the starttime and endtime and the otcreatesched field or element is present ▪ <i>OT before sched</i>: not supported ▪ <i>OT after sched</i>: not supported
Start time	The starttime specified in the import record.
End time	The endtime specified in the import record.
Overtime Length	This field is blank.
Overtime Offset	This field is blank.
Overtime Limit	This field is blank.
Overtime Create sched	If you provide both the starttime and endtime and the otcreatesched field or element is present, this option is enabled.

8.7.10 Deleting Schedule Events & Future Activities

◆ Deleting Scheduled Events

Regardless of the input file format, the **delete** field/element is an optional field. If present, the activity code specified in the import record is replaced with the base schedule code, if the specified activity exists on the schedule.

If the specified activity exists on the agent's schedule during any part of the specified time range, it is deleted and replaced with the base activity code. The time at which the base activity code is placed on the agent's schedule is not necessarily the exact time range specified in the import record.

Note: The **<offexception/>** tag, which is only available for XML input files, indicates to entirely delete a schedule if it exists on the specified date. You can only include the **<delete/>** tag with an **exception** element, and it indicates to only delete the specified activity code. The two tags may not be present in the same schedule record.

◆ Deleting Future Activities

If you run the import program with the **-f** or **-F** command line parameters, future activities are also deleted when a **delete** record is imported.

If a future activity record exists for the agent and the activity code, for the exact time range specified in the import record, it is deleted. If you do not specify a **starttime** in the import record, a future activity record whose start time is not defined is considered a match for the **starttime**; if you do not specify an **endtime** in the import record, a future activity record whose end time is undefined is considered a match for the **endtime**.

It is not necessary for the **otcreatesched** field setting to match the future activity record's *Overtime Create Sched* field setting for the future activity to delete.

If no future activity record matches the import record exactly, no future activity records are deleted.

Future activities are not deleted when you use the **<offexception/>** tag, which is only available for XML input files.

8.7.11 Processing Notes

When an activity code is applied to a schedule using the Import Schedule Changes program, if another activity code exists at the same time on the schedule with the same priority, the program always applies the activity being imported. However, when a future activity record is applied to a schedule, if another activity code exists with the same priority, the activity code with the lower internal ID is applied. Therefore, if schedules are regenerated after the import, there may be a different activity code applied. This is not an issue if activity code priorities are unique.

8.7.12 Lock File

A lock file is placed in the same directory as the input file.

A lock file is maintained for each input file to process records that could not be processed due to locked tables. If not specified on the command line, a default timeout of 120 minutes (2 hours) is used to determine when to quit attempting to process previously locked records. You can specify a different timeout value, in minutes, to use as a command line parameter. The minimum value is 0, which implies that no lock file is maintained and no retries are attempted. The maximum value is 9999 minutes.

The file name format is **original_filename.LCK**. For example, **1627153200.IFC.LCK** indicates the lock file for an input file created at 2:00pm on 7/24/2021 by a system referred to as IFC.

The data format and contents of each record in the lock file is the same as the input record format and contents.

The lock file is in XML format, using the same DTD as the input file.

8.8 Import Schedule Changes Output Files

Regardless of whether you use an XML or pipe-delimited input file, an output file with **.ACK** extension is created in XML format. The output file is placed in the same directory as the input file and is created whether errors occur or not.

You are responsible for removing old output files.

◆ Record Specification

The Import Schedule Changes output file can contain these fields:

Field	Description	Valid Values
id	The ID of the agent. The contents of this field is determined by the -i parameter used on the	If -iA is used:

Field	Description	Valid Values
	program command line. See 8.6 Import Schedule Changes Command Line on page 87 .	1-999999999 If -is used: The value may be 256 alphanumeric characters.
schedule	This is the schedule element; it contains these six child elements.	
schedid	This element contains the schedid field from the import record.	
day	Day of the month on which the schedule occurs.	1-31
month	Month of the year in which the schedule occurs.	1-12
year	The year in which the schedule occurs.	Any year; 4 digits
status	This field indicates the success or failure of the import record processing.	
message	This field contains text providing information about the failure of an import record processing.	

◆ Import Schedule Changes DTD Output File

```
<?xml version="1.0"?>

  <!ELEMENT agentScheduleNotifList (agent+)>
  <!ELEMENT agent (id, schedule)>
  <!ELEMENT schedule (schedid?, day, month, year, status, message*)>

  <!ELEMENT id (#PCDATA)>
  <!ELEMENT schedid (#PCDATA)>
  <!ELEMENT day (#PCDATA)>
  <!ELEMENT month (#PCDATA)>
  <!ELEMENT year (#PCDATA)>
  <!ELEMENT status (#PCDATA)>
  <!ELEMENT message (#PCDATA)>

  <!ATTLIST message errorCode CDATA #REQUIRED>
```

◆ Import Schedule Changes Example .ACK Output File

```
<?xml version="1.0"?>
<agentScheduleNotifList>
  <agent>
    <id>470185</id>
    <schedule>
      <day>12</day>
      <month>8</month>
      <year>2021</year>
      <status>FAILURE</status>
      <message errorCode="4">ERROR: "Lunch" not found for attribute "WB".</message>
    </schedule>
  </agent>
  <agent>
    <id>470185</id>
    <schedule>
      <day>13</day>
      <month>8</month>
      <year>2021</year>
      <status>SUCCESS</status>
    </schedule>
  </agent>
</agentScheduleNotifList>
```

9. Import Vacation Summary

Import Vacation Summary allows you to import agent time off data into the Agent Time Off Summary view such as earned, debited, and carry over, from a standardized file.

Use the Import Vacation Summary feature:

- to reduce the data entry time required to define agents during the initial database build.
- as an automatic feed from a third party system, for example, a human resources database for adding or modifying agents' vacation or PTO balances.

This section covers the following main topics:

- Import Vacation Summary file specifications for importing time off data into the WFM database.
- Input file sample for importing vacation summary data.
- Command line parameters to run the Import Vacation Summary program.
- Import Vacation Summary processing, including information about the output file.
- Import Vacation Summary DTD.

Note: Implementation details to automate the import of time off data are not included.

9.1 Information for the Import Vacation Summary Program

9.1.1 Earning Types

Since earnings types are not included in the import file, the importer adds together each agent's earned values for activity codes that belong to the same earnings type. The total earned value is stored as the earnings type's earned value. This total value overwrites the value in the database.

Example:

Assume **Vacation** and **Personal Time** activity codes both belong to the same earnings type, **PTO**. Agent 123 currently has 48:00 of **PTO**. If the import file includes 40:00 of **Vacation** and 16:00 of **Personal Time** for Agent 123, the agent's **PTO** time is 56:00 after the import.

To import time off information for any code, including **Wait List**, assign the code to an earnings type. Otherwise, the import fails.

9.2 Import Vacation Summary File Specifications

9.2.1 File Name and Location Specifications

To avoid potential problems with processing input/output files, it is recommended you limit the input/output file name to 30 characters or less.

You should put the input file in the specified directory. See [3.1 Import File Name and Location on page 14](#).

9.2.2 Time and Time Zones

Date and time elements referenced in Import Vacation Summary refer to time in the WFM server's time zone.

9.2.3 File Contents – General Specifications

◆ File Elements

The root element for Import Vacation Summary is **agentVacationSummary**.

◆ General Data Requirements

Each **agentVacation** element contains all the time off information to import for a single agent, for one time off group and time off rule. If information is imported for multiple time off groups or time off rules, use separate **agentVacation** elements for the agent.

You can sort records in the input file in any order.

9.2.4 Import Vacation Summary DTD

The file that contains the information to import must follow the DTD shown below. This DTD is the same as the DTD for the XML output of the SmartSync Data Export: Agent Vacation Summary export file.

Note: Do not create a DTD file for this import; only use this as a template for building XML input files.

```
<!ELEMENT agentVacationSummary (agentVacation+)>

<!ELEMENT agentVacation (agent, vacation+)>

<!ELEMENT agent (agentID, vacGroup, vacYear)>
<!ELEMENT vacation (from?, to?, earned?, credited?, debited?, selected?,
taken?, minFullWeeks?, maxPartialHours?, coFrom?, coTo?)>

<!ELEMENT agentID (#PCDATA)>
<!ELEMENT vacGroup (#PCDATA)>
<!ELEMENT vacYear (#PCDATA)>

<!ELEMENT from (day, month, year)>
<!ELEMENT to (day, month, year)>
<!ELEMENT earned (#PCDATA)>
<!ELEMENT debited (#PCDATA)>
<!ELEMENT credited (#PCDATA)>
<!ELEMENT selected (#PCDATA)>
<!ELEMENT taken (#PCDATA)>
<!ELEMENT minFullWeeks (#PCDATA)>
<!ELEMENT maxPartialHours (#PCDATA)>
<!ELEMENT coFrom (#PCDATA)>
<!ELEMENT coTo (#PCDATA)>
```

```

<!ELEMENT day (#PCDATA)>
<!ELEMENT month (#PCDATA)>
<!ELEMENT year (#PCDATA)>

<!ATTLIST vacation type CDATA #REQUIRED>
<!ATTLIST agentID acdID CDATA #IMPLIED>

```

9.2.5 Import Vacation Summary File Contents – Record Specifications

This table includes all elements defined in the Import Agent Vacation Summary DTD. See [9.2.4 Import Vacation Summary DTD on page 102](#). The table consists of the element names, indented based on their level in the DTD, a description of the elements, and the valid values for the elements. If you can edit an element in the WFM user interface, the **Valid Values** column references the corresponding WFM screen.

Element	Description	Valid Values
agentVacationSummary	This is the root element. This element must contain one or more agentVacation children.	
agentVacation	This element contains all the time off information for a single agent for a time off group and time off rule. This element contains a single agent child and one or more vacation children.	
agent	This element must include an agentID , a vacGroup , and a vacYear .	
agentID	The identifier used to find the agent to modify in NICE WFM. This may be a personal ID, a NICE WFM agent ID, an external ID, or an ACD logon ID depending on the -a command line parameter.	If command line parameter is: • -aS - 256 alphanumeric characters • -aT - 1-999999999 • -aA - Any printable ASCII characters except & and \. Maximum length is 64. • -aE - Any printable ASCII characters except & and \. Maximum length is 256. Can be edited on view: <i>Agent Definition</i>
acdID (attribute of agentID)	If you use the -aA command line parameter, you must include this field to define the ACD for which the agent is assigned the logon ID specified in the agentID element above. An ACD must be defined in NICE WFM with this ID. The agent who is assigned the specified logon for the specified ACD on the current date, based on the server time zone, is considered a match.	1-2147483647 Can be edited on view: <i>Agent Definition</i>

Element	Description	Valid Values
	This field is not required if you use the -aS , -aE , or -aT command line parameter.	
vacGroup	The ID of the agent's time off group. A record must exist for the agent in AgentMap whose vac_group_id field matches this ID.	1-2147483647 Can be edited on view: <i>Move Agents</i>
vacYear	The name of time off rule for which the agent's summary information is imported. A record must exist in VacYear whose vac_group_id matches the ID in the vacGroup element, and whose vac_year_name field matches this string. Note: This must be a time off rule with a calculation unit of HH:MM . This import does not support time off rules with a calculation unit of Days .	Any string. A time off rule must be defined with this name for the time off group. Maximum length is 64 Can be edited on view: <i>Time Off Rules</i>
vacation	This element contains all the agent's vacation information for the specified vacGroup and vacYear . This element has a type attribute to indicate the type of time off being imported as well as the following optional child elements: from , to , earned , credited , debited , selected , taken , minFullWeeks , maxPartialHours , coFrom and coTo .	
type	An attribute of the vacation element. This is the description of the activity code for which the VacSummary record is added or updated. If this case sensitive string is Wait List , the VacSummary record with code -1 is updated or added. Otherwise, a record must exist in the Except table whose name, description, matches this case sensitive string and a record must exist in VacType for the specified vacGroup and vacYear for this activity code.	Any printable ASCII characters except and \. Maximum length is 64 Can be edited on view: <i>Time Off Rules</i>
from	The earliest date for which the agent may make requests for this type of time off. This must be a valid date, using day , month , and year children. This date must fall between the time off rule's start and end dates and it must be on or before the specified to date, or the existing record's toDate if you exclude the to element. This date	Can be edited on view: <i>Agent Time Off Summary</i>

Element	Description	Valid Values
	is stored in the fromDate field in VacSummary.	
day	The day of the from .	1-31
month	The month of the from .	1-12
year	The year of the from .	Any year
to	The latest date for which the agent can make requests for this type of time off. This must be a valid date, using day , month , and year children. This date must fall between the time off rule's start and end dates and it must be on or after the specified from date, or the existing record's fromDate if you exclude the from element. This date is stored in the toDate field in VacSummary .	Can be edited on view: <i>Agent Time Off Summary</i>
day	The day of the to .	1-31
month	The month of the to .	1-12
year	The year of the to .	Any year
earned	The amount of time, in minutes, the agent has earned for the time off type in the time off rule date range. This value is stored in the earned field in VacSummary.	0-479520 (999 8-hour days) Can be edited on view: <i>Agent Time Off Summary</i>
credited	The amount of credited time, in minutes for the time off type. This value is stored in the credited field in VacSummary. Note: Credited time is added to the agent's earned time in the Time Off Manger.	0-479520 (999 8-hour days) Can be edited on view: <i>Agent Time Off Summary</i>
debited	The amount of debited time, in minutes for the time off type. This value is stored in the previouslyUsed field in VacSummary. Note: Debited time is counted as taken time in Time Off Manager.	0-479520 (999 8-hour days) Can be edited on view: <i>Agent Time Off Summary</i>
selected	Although this field can be exported using SmartSync Data Export Agent Vacation Summary, it will not be imported, since it is based on Future Activities entered in NICE WFM.	
taken	Although this field can be exported using SmartSync Data Export Agent Vacation Summary, it is not imported,	

Element	Description	Valid Values
	since it is based on schedules in NICE WFM.	
minFullWeeks	The minimum number of weeks the agent is required to take in full week increments.	0-78 Can be edited on view: <i>Agent Time Off Summary</i>
maxPartialHours	The maximum amount of time the agent can take in partial day increments. Note: If you allow agents to take all their time off in partial day increments, this value should be the same as the earned value.	This value must be less than or equal to the record's earned value. Can be edited on view: <i>Agent Time Off Summary</i>
coFrom	The amount of time, in minutes, the agent carried over from the previous year for the time off type. This value is stored in the coFromLastYr field in VacSummary .	0-479520 (999 8-hour days) Can be edited on view: <i>Agent Time Off Summary</i>
coTo	The amount of time, in minutes, the agent carried over into the next year for the time off type. This value is stored in the coToNextYr field in VacSummary .	0-479520 (999 8-hour days) Can be edited on view: <i>Agent Time Off Summary</i>

Note: You can export Agent Vacation Summary information using SmartSync Data Export in the format required for the SmartSync Import Vacation Summary process.

9.3 Import Vacation Summary Input File Example

This is an example of an Import Vacation Summary XML input file.

Example:

```
<?xml version="1.0" encoding="UTF-8" ?>
<agentVacationSummary>
  <agentVacation>
    <agent>
      <agentID>270</agentID>
      <vacGroup>10</vacGroup>
      <vacYear>2020</vacYear>
    </agent>
    <vacation type="PTO">
      <earned>2400</earned>
    </vacation>
    <vacation type="Personal Time">
      <earned>960</earned>
    </vacation>
    <vacation type="Wait List">
      <earned>2400</earned>
    </vacation>
  </agentVacation>
</agentVacationSummary>
```

```

        </vacation>
    </agentVacation>

    <agentVacation>
        <agent>
            <agentID>275</agentID>
            <vacGroup>10</vacGroup>
            <vacYear>2020</vacYear>
        </agent>
        <vacation type="PTO">
            <from><day>1</day><month>1</month><year>2020</year></from>
            <to><day>29</day><month>3</month><year>2021</year></to>
            <earned>2400</earned>
            <credited>480</credited>
            <debited>0</debited>
            <coFrom>1440</coFrom>
            <coTo>0</coTo>
        </vacation>
        <vacation type="Personal Time">
            <from><day>1</day><month>1</month><year>2020</year></from>
            <to><day>29</day><month>3</month><year>2021</year></to>
            <earned>0</earned>
            <credited>0</credited>
            <debited>0</debited>
            <coFrom>0</coFrom>
            <coTo>0</coTo>
        </vacation>
        <vacation type="Wait List">
            <from><day>1</day><month>1</month><year>2020</year></from>
            <to><day>29</day><month>3</month><year>2021</year></to>
            <earned>3360</earned>
            <credited>0</credited>
            <debited>0</debited>
            <coFrom>0</coFrom>
            <coTo>0</coTo>
        </vacation>
    </agentVacation>
</agentVacationSummary>

```

9.4 Import Vacation Summary Command Line

For distributed systems, including hosted/cloud environments, copy the import file onto the EFT site before executing the import vacation summary command. A process automatically moves the file from the EFT site to the server. It can take SmartSync 10 minutes or more to process files depending on the size of the files and the timing of tasks to move files.

You can execute the program manually or automatically on a pre-determined schedule.

Note: The Import Vacation Summary input file is not deleted after a successful import. You are responsible for deleting the import file.

9.4.1 Import Vacation Summary Command Line Parameters

Import Vacation Summary is invoked using the `importVacSummary` command and the following parameters:

Parameter	Description
-c<custID> (required)	The ID of the customer partition for which the data is imported. 1 – 32757
-f<path/file.name> (recommended)	Use this parameter to specify the path and name of the file to import. The recommended path and file name is <code>/totalview/ftp/imports/customer<custID>/vacation/vacationImport.xml</code> If you do not provide a file parameter on the command line, the default path and file name is <code>/totalview/ftp/imports/vacSummary/vacSummary.xml</code> .
-a<ID type> (optional)	The criterion used to match agents in the XML file with agents in the WFM database. The possible values, not case sensitive, are: ▪ T or t – Use the WFM agent ID to identify agents. Values specified in agentID elements, see agentID on page 103, must be WFM agent IDs. ▪ S or s – Use the personal ID to identify agents. Values specified in agentID elements, see agentID on page 103, must be values stored in the Personal ID field in the agent definition. Values must be unique personal identification numbers. ▪ E or e – Use the external ID to identify agents. Values specified in agentID elements, see agentID on page 103, must be values stored in the External ID field in the agent definition. ▪ A or a – Use the ACD login ID to identify agents. Values specified in agentID elements, see agentID on page 103, must include an acdID attribute and the agentID element contains the associated login ID. For example, to identify the agent whose login ID on ACD 25 is 1234, the id element is: <code><agentID acdID="25">1234</agentID></code> If you do not include this parameter, the program uses -aT to identify agents. Note: If you use ACD logon to identify agents, the agent assigned the specified logon for the specified ACD on the current date, based on the server time zone, is considered a match.

9.4.2 Import Vacation Summary Command Line Example

Refer to the *Administrator Guide - WFM* for details about running commands from the Custom Activities view.

Linux:

```
importVacSummary -c<custID> -  
f/totalview/ftp/imports/customer<custID>/vacation/vacationImport.xml -aT >  
/totalview/ftp/imports/customer<custID>/vacation/run.log
```

Windows:

```
importVacSummary -c<custID> -  
f/totalview/ftp/imports/customer<custID>/vacation/vacationImport.xml -aT >  
/totalview/ftp/imports/customer<custID>/vacation/run.log 2>&1
```

9.5 Import Vacation Summary Processing

Processing ends and an error message prints indicating the reason for the failure in these circumstances:

- you omitted the customer command line parameter, **-c**, or if the specified customer ID is invalid. The message “The customer ID is invalid.” prints.

The customer ID is considered invalid if either of these conditions is true:

- you did not specify a valid integer
- there is no customer in NICE WFM with the id specified
- the file specified with the **-f** command line parameter does not exist
- you omitted the **-f** command line parameter and the default path and file name, `/totalview//ftp/imports/vacSummary/vacSummary.xml`, does not exist
- you included the agent identifier command line parameter, **-a**, but used something other than **S, s, A, a, E, e, T** or **t**

If an element contains a blank value or if you omit an element which correlates to a required database field in a default record, the existing value in the WFM database which corresponds to the element is cleared.

If no record exists in the **VacSummary** table for the specified agent, time off group, time off rule, and time off type, a new record is created.

If a record already exists for the specified agent, time off group, time off rule, and time off type, it is modified. If you omit a child of the vacation element, the corresponding database field is not modified.

Note: The **vacYear** element must be a time off rule with a calculation unit of **HH:MM**. This import does not support time off rules with a calculation unit of **Days**.

9.6 Import Vacation Summary Output File

An output file is created in the same directory as the import file. The name of the output file is the same as the import file, with the **.err** extension. For example, **input.xml.err**.

The output file contains the specified customer ID, the path and name of the import file, the date and time processing started and ended, and any errors that occurred.

Example .ERR file:

```
<!-- Customer ID: 1 -->
<!-- Import file: input.xml -->
<!-- Start time: 8/17/22 10:27 PM -->
<!-- End time: 8/17/22 10:27 PM -->

<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<agentVacationSummary>
  <agentVacation>
```

```
<messages>Error matching agent by WFM ID: 100 .</messages>
<agent>
  <agentID>100 </agentID>
  <vacGroup>1</vacGroup>
  <vacYear>2</vacYear>
</agent>
  <vacation type="a"/>
</agentVacation>
</agentVacationSummary>
<!-- AgentVacation elements -->
<!-- 1 AgentVacation elements were processed -->
<!-- 1 errors were found -->
<!-- 0 AgentVacation elements were successfully imported -->
<!-- 1 AgentVacation elements were not imported -->
```